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Hydrogel-Based Flexible Sensors: From Manufacturing Strategies to Sensing Applications

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Hydrogel-Based Flexible Sensors: From Manufacturing Strategies to Sensing Applications

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Hydrogel-based flexible sensors: from manufacturing strategies to sensing applications

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Abstract

Hydrogel-based flexible sensors combine tissue-like mechanical properties with superior sensing performance, showing immense potential in human-machine interactions, soft robotics, and health monitoring. Although significant progress has been made in related research, the fabrication of precise structures, the environmental tolerance of materials, and the heterogeneous integration between materials and devices remain key research challenges. In response to these challenges, this review provides a comprehensive review of the latest progress in hydrogel-based sensors, covering the entire process from material fabrication and functional regulation to interface engineering. First, the impacts of advanced processing techniques such as solution processing, microstructure molding, photolithography, additive manufacturing, and fiber spinning on film structure are discussed in detail. Subsequently, physicochemical strategies for enhancing sensing performance and environmental tolerance are analyzed. Furthermore, diverse assembly schemes and interface engineering strategies are highlighted. Finally, typical applications in pressure, strain, environmental, and physiological signal monitoring are summarized. This review aims to provide guidance for the development of next-generation high-performance and stable flexible electronic systems.

Keywords: hydrogels, manufacturing process, functional regulation, assembly, flexible sensors

Highlights

1. Conducts in-depth exploration of hydrogel-based sensors from manufacturing and performance optimization to assembly and application.
2. Analyses typical hydrogel fabrication techniques and corresponding performance regulation strategies.
3. Highlights the strategic assembly designs for hydrogel-based sensors.
4. Summarizes the applications of hydrogel-based sensors in flexible electronics.
5. Outlines current challenges and future development trends of hydrogel-based sensors.

1. Introduction

With the rapid advancement of the internet of things (IoT), artificial intelligence (AI), and personalised medicine, modern information technology is undergoing a paradigm shift from rigid devices to flexible electronics. In cutting-edge fields such as human–machine interaction (HMI)^[1], soft robotics^[2], and health monitoring^[3], developing high-performance sensing materials that seamlessly conform to dynamic skin and exhibit tissue-like mechanical properties has become pivotal for bridging the digital world with biological systems. Owing to their unique physicochemical structure, hydrogels have emerged as the core material for constructing next-generation flexible interfaces^[4, 5].

Hydrogels, defined as three-dimensional (3D) hydrophilic polymer networks, are widely present in natural systems, including plants, animals, and microorganisms^[6, 7]. The developmental timeline of the hydrogels is illustrated in Figure 1. The term “hydrogel” was first proposed in 1894 by Van Bemmelen, initiating scientific interest in this material^[8]. After decades of research, Wichterle and Lim successfully synthesized poly(2-hydroxyethyl methacrylate) hydrogels in 1960 and developed the first biocompatible soft contact lenses based on this material, marking a significant advancement in hydrogel research and applications^[9]. This breakthrough signifies a shift in research focus from material exploration to performance optimization, thereby facilitating the development of hydrogels with conductivity, stimulus-responsive properties, and adhesive capabilities^[10–15]. These materials demonstrate biomimetic flexibility and permeability comparable to those of biological tissues, showing remarkable potential in biomedical applications such as wound healing^[16], drug release^[17], and tissue engineering^[18, 19]. With advancements in materials science and flexible electronics, hydrogel applications have expanded to include sensors^[19–29], supercapacitors^[13, 30, 31], and bioelectrodes^[32, 33]. Such mechanical compliance and sensitivity to external stimuli (e.g., pressure, strain, and temperature) enable innovative solutions for health monitoring and human–machine interfaces. Post-2020, hydrogel development entered a new phase characterized by interdisciplinary integration and AI, facilitating the emergence of hydrogels in cutting-edge fields, including smart healthcare^[34–37], bionic electronic skin^[38–43], and soft robotics^[44–47].

Currently, numerous reviews have been published concerning hydrogels in the field of flexible electronics. For instance, a review on hydrogel-based flexible electronics was presented^[48]. This review primarily focused on the latest synthetic methods for advanced functional hydrogels and corresponding cutting-edge applications across various domains, with a particular emphasis on the correlation between

hydrogel properties and device performance. Similarly, the application of ionic hydrogel sensors in next-generation personalized medicine has been comprehensively reviewed^[49]. This review systematically organised information from material sources, conductive mechanisms, preparation strategies, and key characteristics. Regarding environmental adaptability, a systematic overview was provided on hydrogels for extreme environments, where strategies for achieving anti-mechanical damage, anti-detachment, anti-swelling, anti-freezing, and anti-foreign body responses were summarized from material development and structural design perspectives^[50]. Meanwhile, significant attention has been given to multifunctional conductive hydrogel sensors for wearable applications, and advances in strain/pressure sensors, sweat sampling, and touch panels enabled by hydrogels with self-healing, self-adhesive, and anti-freezing capabilities have been systematically elaborated upon in another review^[51]. These reviews offer valuable insights into understanding the chemical properties and functional development of hydrogels. However, a closer examination of the current state of the field reveals that translating hydrogels from laboratory prototypes into high-performance devices still faces significant challenges at the levels of advanced manufacturing and device integration. This constitutes the core motivation for this review. First, the bulk synthesis of hydrogels via simple cast-film methods alone struggles to meet the demand for the precise micro- and nano-scale structural control required by next-generation sensors. Second, synergistically optimising synthetic strategies and encapsulation techniques to confer structural integrity and functional stability across diverse environments remains an urgent manufacturing challenge^[52]. Moreover, the fabrication of flexible sensors inherently involves a cross-scale heterogeneous integration process. The unique 'soft and wet' properties of hydrogels result in significant mechanical mismatches with rigid electrodes and encapsulation layers^[53, 54]. The establishment of robust heterogeneous interfaces during manufacturing to prevent delamination and failure under dynamic loads represents a major technical bottleneck in current device processing.

This review comprehensively explores the latest research advances in hydrogel-based flexible sensors and systematically investigates hydrogel synthesis techniques, functional tuning strategies, sensor assembly processes, application domains, as well as current challenges and future development trends (see Figure 2). The structure of this paper is as follows: section 2 provides a comprehensive overview of advanced manufacturing techniques for hydrogels, encompassing solution processing, microstructure molding, photolithography, additive manufacturing, and fiber spinning, which serve as the cornerstone for achieving customised device structures. Section 3 introduces performance-tuning strategies for hydrogels, systematically

examining approaches to constructing conductive networks, designing for extreme environmental adaptability, implementing self-healing mechanisms, achieving biocompatibility, and imparting antimicrobial properties. Section 4 summarizes structural design principles for hydrogel-based sensors, focusing on heterogeneous interface connection strategies and device encapsulation solutions. Section 5 outlines the applications of hydrogel-based sensors in monitoring pressure, strain, environment, and bioelectrical signals, demonstrating their immense potential in flexible electronics and intelligent systems. Finally, Section 6 analyses unresolved key issues and technical bottlenecks in current hydrogel-based sensor research while outlining future development directions. This review aims to provide systematic theoretical underpinnings for the preparation of high-performance hydrogels and the design and fabrication of sensors. Concurrently, it is hoped that this work provides a reference framework for in-depth interdisciplinary research in this field and for engineering applications of flexible electronic devices.

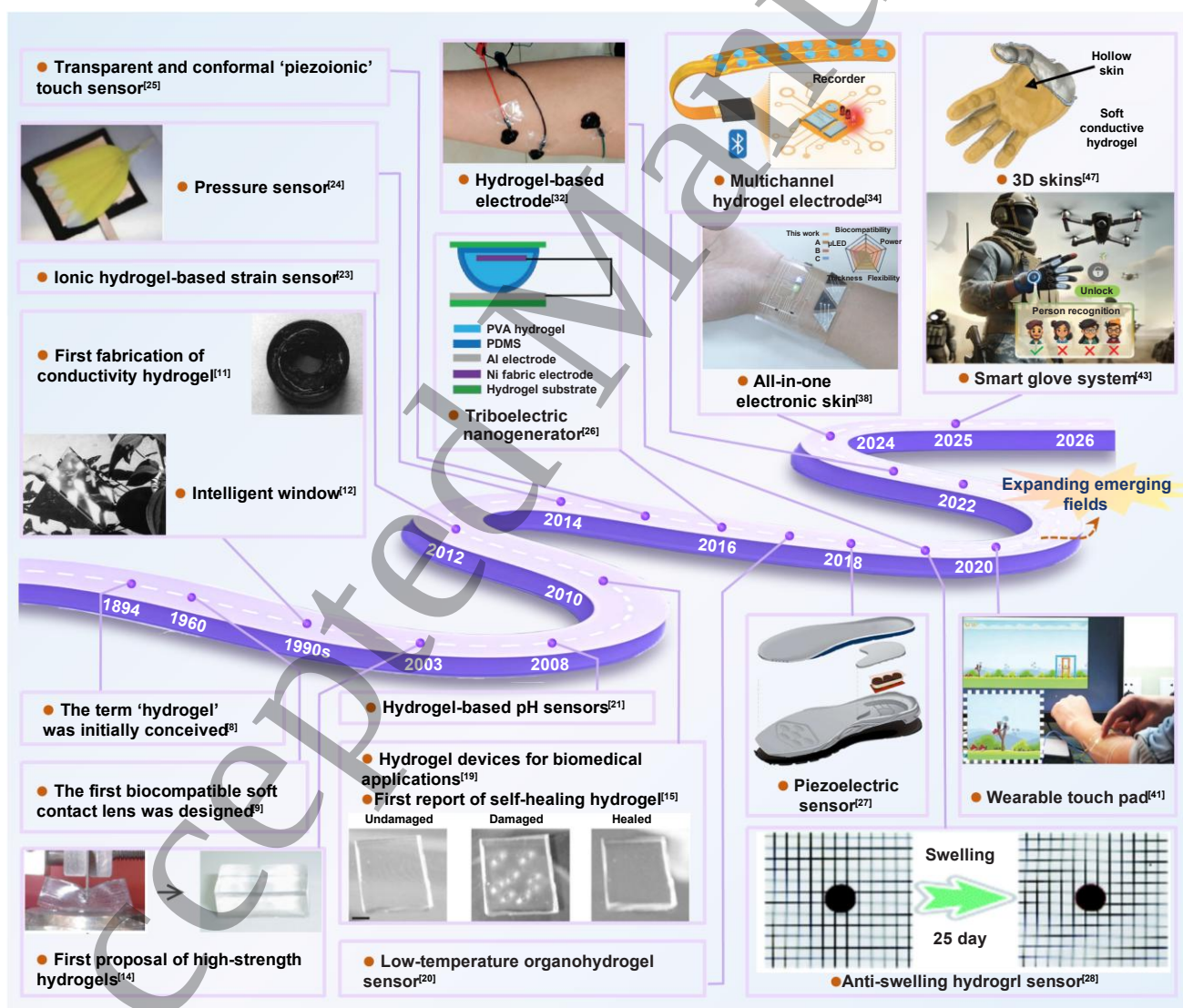


Figure 1. The timeline of the development of hydrogels and hydrogel-based sensors. Image for “first fabrication of

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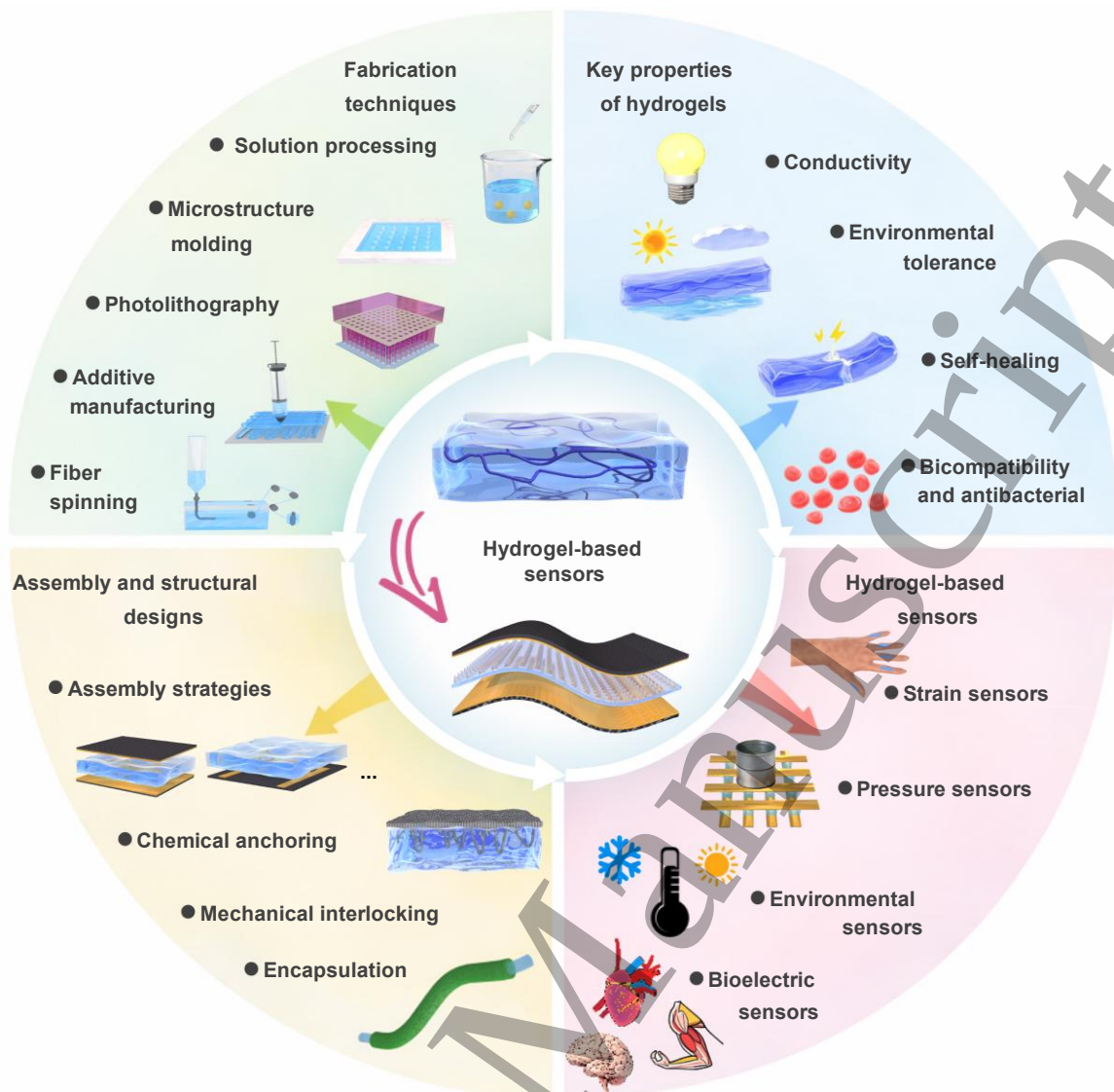


Figure 2. Schematic overview of the review framework. It systematically covers the fabrication and performance regulation strategies of hydrogels, along with the assembly, structural design, and applications of flexible hydrogel-based sensors.

2. Fabrication techniques for hydrogels

Previous gelation strategies have established foundational physicochemical principles for hydrogel formation, yet traditional bulk synthesis alone cannot meet the complex demands of next-generation flexible electronics. In human-machine interfaces and soft robotics, devices must achieve seamless conformal contact with irregular biological tissues, provide precise mechanical feedback, and adapt their morphology in dynamic environments. These advanced capabilities require multi-scale structural engineering beyond simple gelation control. The fabrication techniques define not only hydrogel shape accuracy but also pore architecture and orientation to determine device sensitivity, linearity, and anisotropic response. This section systematically reviews key hydrogel structural design paradigms from a manufacturing perspective, covering solution processing, microstructure molding, photolithography, additive manufacturing, and fiber spinning.

2.1. Solution processing

Solution processing is the most fundamental and versatile strategy for hydrogel preparation. Its core process involves precursor composition regulation and liquid-solid transition. To introduce functional components and regulate formulations, four main technical routes exist: physical casting, one-pot in-situ polymerization, solvent exchange, and dip-coating.

Physical casting is the most intuitive method with the lowest technical barrier. It is primarily suitable for pre-synthesized water-soluble high molecular weight polymers, such as polyvinyl alcohol (PVA), gelatin, sodium alginate (SA), or chitosan (CS)^[55, 56]. This process involves dissolving polymer powder in a solvent (e.g., water, organic solvents, etc.) to form a homogeneous solution, which is then cast onto a mold or substrate. Subsequently, physical methods like freeze-thaw or cooling induce a liquid-solid transition for shaping (Figure 3(a)). In contrast to casting methods based on macromolecular physical assembly, one-pot in-situ polymerization allows for the direct construction of crosslinked networks from a precursor mixture of monomers, crosslinkers, initiators, and solvents. As shown in Figure 3(b), a single step of irradiation or thermal treatment triggers free radical polymerization, thereby inducing rapid solidification and obviating the need for complex post-processing^[57, 58]. This process not only precisely replicates micro/nano textures from molds while maintaining high coating flatness but also ensures the uniformity of the polymer network at the microscopic scale^[59]. Based on the principles of in-situ polymerization, the HSCP hydrogel was fabricated by ultraviolet (UV) irradiation of a mixture containing acrylamide (AM), honey, NaCl, the crosslinker N,N'-methylene diacrylamide (MBA), and a photoinitiator for 45 min, yielding a network with excellent self-

adhesion, conductivity, and antibacterial properties^[60]. When monomers are water-insoluble or specific functional assemblies are needed, the solvent exchange strategy serves as a versatile method for structural shaping and functionalization. A typical strategy involves the initial formation of an organogel network in an organic solvent, followed by immersion of the resultant gel in water. During this stage, a concentration gradient drives solvent exchange, wherein the progressive displacement of the organic solvent by water induces phase separation or reorganization of the polymer chains, ultimately yielding a hydrogel (Figure 3(c))^[61, 62]. Additionally, by immersing the preformed hydrogel in a solution containing functional components, solvent permeation can fill these components into the interstitial spaces of the polymer network^[63–65]. Such a diffusion-based post-processing method can avoid the inhibition of polymerization reactions by strong oxidants or high-concentration ions. For example, a smart adaptive ZL-PAM hydrogel was prepared by soaking pre-formed PAM hydrogel in ZnSO₄ and LiBr solutions to introduce Zn²⁺ and Li⁺. Hygroscopic Li⁺ provides excellent moisturizing ability, while Zn²⁺ suppresses byproducts (Zn(OH)₂) and modulates water content. It enables reversible switching between water-rich and water-deficient states for crosstalk-free dual sensing of oxygen and humidity in a single device^[64]. Similarly, a PAM/Ca-Alg double-network hydrogel fiber was also fabricated via a soaking strategy. The fibers were sequentially immersed in CaCl₂ and LiBr solutions to form an ionically crosslinked network and introduce hygroscopic Li⁺ and Br⁻ ions, achieving excellent mechanical properties and drying and freezing tolerance^[65]. As a complement to the aforementioned strategies, dip-coating offers a pathway for the fabrication of thin and integrated hydrogels by incorporating prefabricated scaffolds. This method involves immersing high-porosity nanonetworks (such as nanofiber membranes or porous scaffolds) into a hydrogel precursor solution, utilizing micro/nanoscale capillary effects to drive rapid infiltration and wetting of the framework. Subsequently, through appropriate physical or chemical curing processes, a morphologically uniform and ultrathin hydrogel is formed under the physical confinement of the nanoscaffold (Figure 3(d))^[66]. Specifically, through this approach, organohydrogel nanofilms with a thickness of only 392 nm were prepared by infiltrating polyurethane (PU) nanonetworks with a gelatin-glycerol (Gly) precursor solution^[67]. These nanofilm electrodes exhibited exceptional skin conformability and environmental stability, enabling continuous and stable electrocardiogram (ECG) monitoring for up to 9 days. This approach not only overcomes the thickness control limitations inherent in traditional casting methods but also leverages the mechanical reinforcement of the nanonetwork to enhance the structural integrity of the film, resulting in hydrogels with superior flexibility and ultrathin characteristics.

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Once the precursor is prepared, fluid processing techniques convert it into specific geometries for designated applications. Conventional casting enables tunable hydrogel thickness from micrometers to millimeters via dimensionally customized molds (Figure 4(a)). Spin coating uses centrifugal force to spread the precursors into uniform microscale films (Figure 4(b)). Spray coating employs high-pressure atomization for directional deposition, making it ideal for creating functional coatings on substrates (Figure 4(c)). Regarding spin-coating applications, hydrogel thin films with controllable thicknesses have been successfully fabricated^[68]. A polyacrylamide (PAM)/carrageenan hydrogel film with tunable thickness (54.61–85.72 μm) was fabricated by spin-coating the precursor onto oxygen plasma-treated polydimethylsiloxane (PDMS) substrates at speeds of 250–1 000 rpm. In another approach, a modified spin-coating strategy was employed to fabricate an ionogel-based humidity sensor array, producing uniform sensing films with thicknesses down to 2.4 μm and achieving ultrafast response/recovery times of 0.65 s/0.85 s^[69]. Spray coating not only is widely used for the rapid prototyping of hydrogel wound dressings and the construction of surface conductive layers, but also serves as an important method for fabricating high-performance gel thin films. Further investigations have demonstrated the efficacy of spray coating in developing flexible sensing interfaces^[70]. In this approach, a mixture of polymer matrices and functional solvents is atomized and deposited onto heated substrates. Precise regulation of the spray pressure and duration facilitates the rapid formation of uniform and dense gel networks. Sensors fabricated via this process retain the advantages of simple operation and easy patterning.

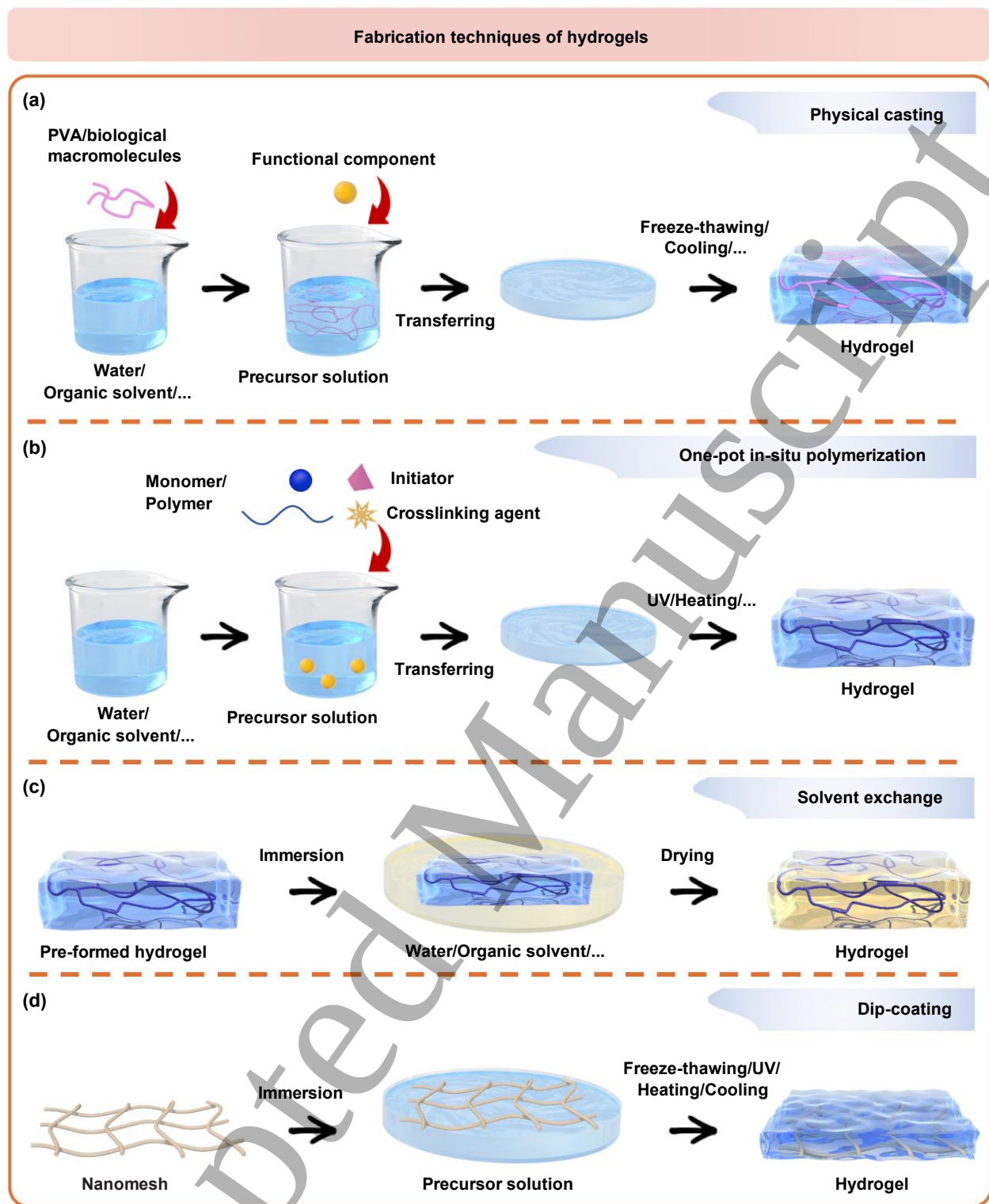


Figure 3. Various solution processing techniques have been developed for hydrogel fabrication, as illustrated below: (a) physical casting, (b) one-pot in-situ polymerization, (c) solvent exchange, (d) dip-coating.

2.2. Microstructure molding and photolithography

To create specific topological structures on hydrogel surfaces for enhanced sensing performance, molding and photolithography are currently the most widely used surface micromachining techniques. Molding primarily

relies on pre-fabricated master templates, which can be precision-made via lithographic silicon wafers or laser etching, or directly replicated from textured materials. The typical process involves casting a formulated hydrogel precursor onto the template surface. After physical or chemical gelation, demolding transfers the template's topographic features with high fidelity to the hydrogel surface. The essence of such microstructural design lies in altering the mechanical conduction path of the device. Structures such as micro-pyramids, micropillar arrays, or interlocked geometries exhibit nonlinear changes in contact area and generate localized stress concentrations under compression^[71]. This geometric enhancement enables sensors to produce significant signal variations even under extremely low pressure, effectively overcoming the sensitivity limitations of conventional flat films. As shown in Figure 4(d), an AM/agarose/NaCl ionic hydrogel was fabricated using a commercial anti-micropyramid template. The micro-pyramid structure significantly enhanced the capacitive sensing performance through stress concentration, achieving a high sensitivity of 172 kPa⁻¹ within 30 kPa, a fast response time of 11.2 ms, and a low detection limit of 0.5 Pa^[72]. Furthermore, to reduce process costs and optimize demolding, secondary transfer strategies based on soft molds are widely adopted. Researchers often use low-surface-energy silicone elastomers such as PDMS or Ecoflex to replicate precision photolithographic silicon wafers or naturally textured biological surfaces into flexible intermediate molds. Hydrogel precursors are then cast into these flexible molds^[73]. This approach not only protects delicate master templates from mechanical damage during repeated operations but also leverages the inherent low surface energy and excellent elastic deformation of soft molds to significantly reduce interfacial adhesion and facilitate demolding after gelation. Consequently, it ensures microstructural integrity while enabling efficient mold reuse. In contrast, photolithography utilizes photochemical reactions to achieve higher-resolution patterning, employing photoinitiators sensitive to specific wavelengths and photomasks to regulate the spatial distribution of UV light. Exposed regions undergo crosslinking and solidification, while unexposed areas remain liquid and are washed away, thereby constructing high-density micro-nano structures^[74]. A representative strategy utilized direct microlithography for the fabrication of high-density gel-skins. The key innovation lay in creating a photosensitive gel material and realizing a patterning process without photoresist. By employing a mask to selectively expose and develop the gel precursor under UV light, high-precision gel arrays were directly produced, which significantly streamlined the complex procedures of traditional photolithography (Figure 4(e))^[75]. Such a strategy successfully produced high-density flexible electronic skin, offering new insights for efficient device integration.

2.3. Additive manufacturing

Additive manufacturing technology fabricates complex 3D hydrogel structures by digital model control and layer-by-layer stacking, greatly expanding design freedom. Among these, direct ink writing (DIW) printing, which is the most widely used extrusion-based technique, relies on micro-nozzles to extrude hydrogel inks with tunable rheological properties. Recent advances reveal that the high shear flow field inside the nozzle can induce the alignment of anisotropic particles or nanoreinforcing phases along the printing path^[76, 77]. For instance, a composite ink containing conductive ceramic flakes (~200 nm thick) was extruded through a 35 mm-long cylindrical nozzle^[78]. Shear forces during printing induced highly oriented alignment of the flake fillers, yielding composite organohydrogel monofilaments with concentric layered microstructures. These monofilaments were further printed into biomimetic macroscopic architectures such as unidirectional arrays, endowing the material with significant anisotropy (Figure 4(f)). Notably, controlling the surface tension of hydrogel ink is crucial; excessive surface tension can cause droplet splashing or satellite droplet formation during jetting, severely degrading the edge sharpness and resolution of printed patterns. As a representative of photocuring, digital light processing (DLP) printing offers higher forming accuracy (~20–250 μm) than DIW printing does^[79, 80]. This technology uses digital projection to cure photosensitive precursors layer by layer via area exposure, enabling internal interconnected micropores or overhanging structures that are difficult to achieve with extrusion-based methods. DLP printing materials typically consist of photoactive polymers, photoinitiators, and photoabsorbers^[81]. By adjusting the content and ratio of the first two components, the mechanical properties of DLP-printed hydrogels can be tailored. The photoabsorber also plays a critical role, as it influences the printing resolution during the photopolymerization process^[82, 83]. For instance, DLP printing was utilized to fabricate organohydrogels with tailored architectures, such as diamond, grid, and peanut-like geometries^[84]. The resulting ionic and electronic hydrogels exhibited remarkable stability, yielding gauge factors of 4.5 (0%–200% strain) and $13.5/1.5 \times 10^6$ (0%–2%/2%–100% strain), respectively. A 3D grid structure sensor printed and integrated into an HMI system achieved precise control with a minimum monitoring angle of 5° (Figure 4(g)). To further improve the microstructural resolution of sensors, two-photon polymerization employs femtosecond lasers based on nonlinear absorption. This technique achieves feature sizes from the micrometer scale down to below 100 nm, albeit with long fabrication times (hours) and limited construction dimensions. For example, researchers have used this technique to fabricate photonic crystal hydrogel microsensors within microfluidic chips. The sensors achieve submicron resolution (16 μm diameter)

and enable in-situ, real-time monitoring of pH and metal ions^[85]. Expanding on additive manufacturing, four-dimensional (4D) printing incorporates the dimension of time. By integrating smart materials like hydrogels, printed structures can dynamically evolve in shape or function upon exposure to triggers such as temperature, light, or chemicals. Leveraging the properties of smart materials and specific structural designs, such as heterogeneous network architectures and spatial stress release mechanisms, the process transforms simple geometries into complex 3D configurations, transcending the static constraints of traditional manufacturing to offer superior dynamic adaptability^[86]. Specifically, researchers have fabricated hydrogel films via a single-step digital exposure process, achieving trigger-free shape shifting with predefined onset periods. Such a methodology enables the rapid construction of permanent 3D structures and spatial control over shape-recovery timing by modulating exposure duration. Notably, 4D-printed self-wrapped hydrogel electrodes achieved conformal contact with rabbit extraocular muscles to successfully record action potentials under various stimulus currents^[87]. The progress enhances the implant adaptability of bioelectronic devices, presenting a new paradigm for highly adaptive intelligent bionic systems.

2.4. Fiber spinning

Processing hydrogels into one-dimensional continuous fibers is key for wearable smart textile integration. Wet spinning, as the process for hydrogel fiber production, involves extruding a polymer dope through a spinneret into a coagulation bath. During this process, the extruded jet undergoes diffusion or rapid ionic crosslinking, inducing instantaneous phase separation and solidification into nascent fibers. Subsequent mechanical drawing further promotes molecular chain alignment along the fiber axis, enhancing mechanical strength^[88, 89]. As illustrated in Figure 4(h), bioinspired aligned hydrogels were fabricated via a wet-spinning strategy. This process involved injecting a mixed solution of PVA, polydopamine (PDA)-carbon nanotubes (CNTs), and poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT: PSS) into an ammonium sulfate coagulation bath. The shear flow generated during extrusion facilitated the alignment of the polymer chains, resulting in the formation of fibrous filaments. These fibers were then immersed in a sodium pyrrolidone carboxylic acid (PCA-Na) solution, where salt ions promoted interchain aggregation and crystallization, effectively welding individual fibers into a structurally anisotropic bulk hydrogel^[90]. As a refined evolution of wet spinning, microfluidic spinning enables precise manipulation of multiphase fluids by leveraging laminar flow characteristics within microchannels, enabling the fabrication of hydrogel fibers with core-shell

cross-sections^[91]. For instance, a novel hydrogel-assisted microfluidic spinning technique circumvents the limitations of conventional spinning in processing difficult-to-spin polymers by encapsulating prepolymers within protective hydrogel fibers of arbitrary length (Figure 4(i))^[92]. The approach also enables the reshaping of uncured prepolymer fibers to produce helical fibers with excellent stretchability and mechanical compliance. Additionally, electrospinning uses a high-voltage electrostatic field to stretch polymer solutions into micro-nano jets, which accumulate on a collector to form nanofiber membranes with high porosity and specific surface area. The process is ideal for fabricating breathable epidermal electrodes^[93]. Despite its potential, direct electrospinning of pure hydrogel fibers remains restricted. Such a limitation arises from the high water content and low viscosity of the hydrogel precursor systems, which hinder stable jet stretching and often lead to electrospraying instead of continuous fiber formation.

To provide a systematic overview of the fabrication techniques discussed above, Table 1 summarizes the key characteristics of each method, including extreme precision, fabrication efficiency, key advantages, primary limitations, and their implications for sensor performance. This comparative analysis aims to offer guidance for selecting appropriate fabrication strategies based on specific application requirements.

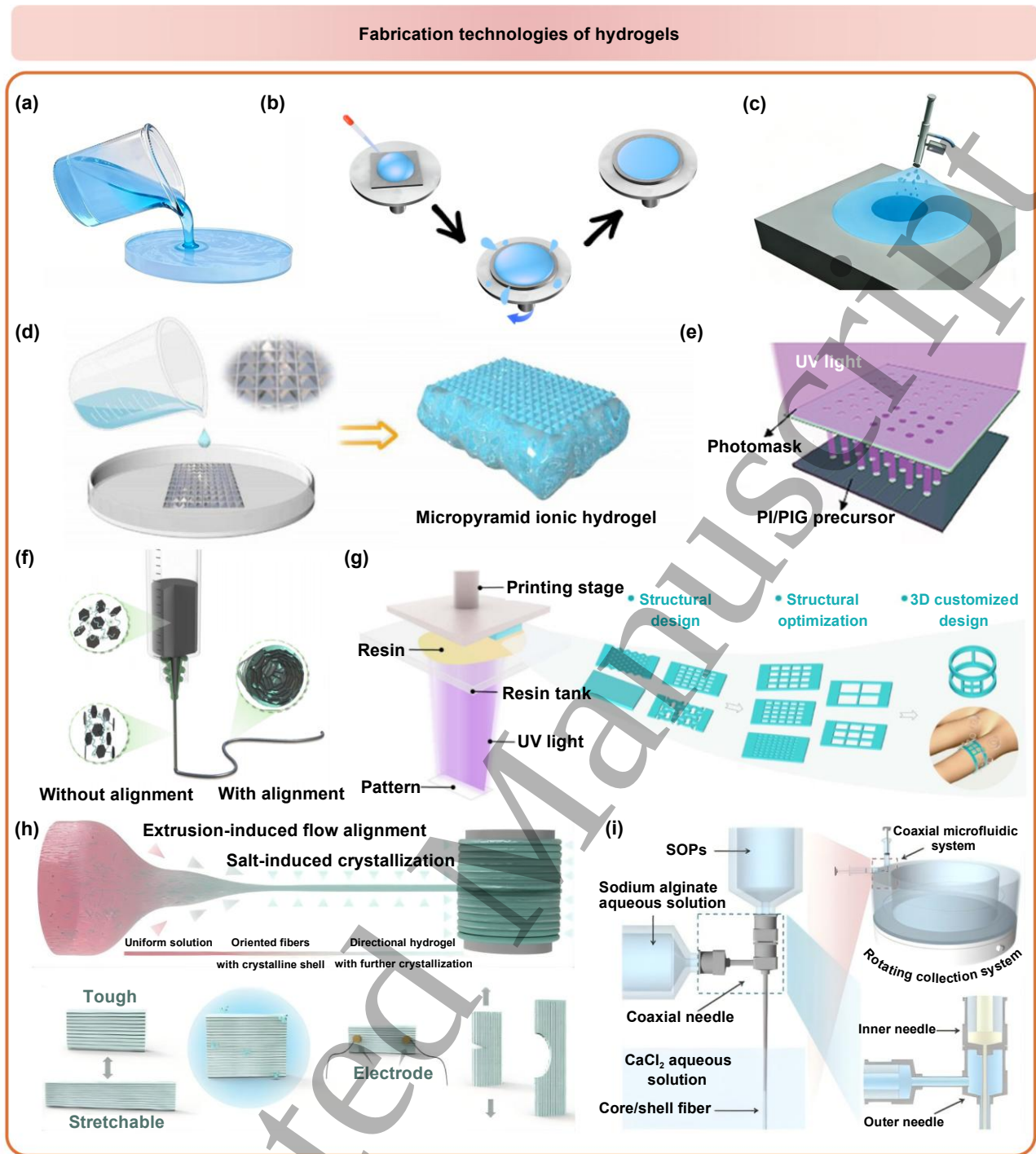


Figure 4. Hydrogel fabrication techniques. (a) Casting. (b) Spin coating. (c) Spray coating. (d) Ionic hydrogel cast in a pyramid-shaped mould. Reproduced from [72]. CC BY 4.0. (e) Fabrication of high-precision gel arrays by direct microlithography. [75] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (f) Constructing anisotropic organohydrogel filaments by shear-induced alignment in DIW 3D printing. Reproduced from [78]. CC BY 4.0. (g) DLP 3D-printed organohydrogels with tailored architectures. Reproduced from [84]. The Author(s). CC BY 4.0. (h) Wet-spinning strategy for constructing bioinspired aligned hydrogels. Reproduced from [90]. CC BY 4.0. (i) Hydrogel-assisted microfluidic spinning technology. Reproduced with permission from [92]. Copyright © 2023 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S.

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Table 1. Comparison of representative hydrogel fabrication techniques.

Fabrication techniques	Extreme precision	Fabrication efficiency	Key advantages	Primary limitations	Implications for sensor performance	Refs.
Dip coating	Nano-scale (~400 nm)	High (suitable for large-area fabrication)	Enables ultrathin hydrogels when combined with nanonetworks; capillary-driven infiltration; tunable thickness	Geometry-limited; susceptible to swelling and stability issues; limited control over shape and dimensions	Realizes ultrathin, conformable skin electrodes; enhances long-term monitoring stability	[65–67]
Casting	Micron-scale (≥50 μm)	High (suitable for large-area fabrication)	Simple operation, low cost, suitable for large-area preparation; no complex equipment required	Limited template precision; difficult thickness control	Uniform intrinsic properties but limited spatial resolution; suitable for basic sensing substrates	[94–96]
Spin coating	Micron-scale (<5 μm)	High (suitable for large-area fabrication)	Excellent thickness uniformity, tunable thickness	Limited device space; constrained by stringent solution requirements: viscosity, concentration, and film-forming properties	Suitable for high-sensitivity ultrathin sensing layers	[68, 69, 94, 97]
Spray coating	Micron-scale (10 μm)	High (suitable for large-area fabrication)	Conformal coating on complex surfaces; suitable for irregular substrates	Difficult thickness uniformity control; material waste; requires optimized atomization parameters	Suitable for wearable sensors on irregular substrates; can construct porous sensitive structures	[98–101]
Microstructure molding	Submicron to hundreds of micrometers	High (reusable soft mold, batch replication)	High-fidelity replication of microstructures; soft mold facilitates demolding; protects master template; Elastomeric mold secondary replication reduces process cost	Template fabrication requires lithography or laser etching; high cost of precision molds; demolding may damage the structure; high aspect ratio structures prone to collapse	Significantly enhances sensitivity through stress concentration effects; microstructures modulate signal response	[5, 72, 102]
Photolithography	Micron-scale (~7 μm)	Relatively high (mask-dependent)	High resolution; excellent reproducibility; enables high-density arrays	Expensive equipment; limited to 2D patterns; solvents may damage hydrogels	Realizes highly uniform microscale sensor arrays; suitable for integrated sensing systems	[75, 103]

Table 1. Continued.

Fabrication techniques	Extreme precision	Fabrication efficiency	Key advantages	Primary limitations	Implications for sensor performance	Refs.
DIW printing	Micron-scale (nozzle)	Medium (dependent on layer height)	Multi-material capability; shear-induced filler alignment; constructs anisotropic structures	Limited resolution; nozzle clogging; critical surface tension control	Realizes anisotropic sensors; shear alignment enhances directional sensitivity	[104–106]
DLP printing	Micron-scale (~20–250 μm)	Relatively high (dependent on layer height and structural complexity)	High resolution, rapid fabrication of complex 3D structures; can construct suspended/interconnected porous structures	Limited material selection (photocurable resins only); requires photoabsorber to control penetration depth	Realizes high-resolution 3D sensor structures; customized geometries modulate response	[81, 107, 108]
Two-photon polymerization	Nano-scale (~100 nm)	Extremely low (fabricating microstructures)	Sub-100 nm resolution; 3D nanofabrication; breaks diffraction limit	Extremely slow, high cost; limited build volume	Realizes 3D nanosensors and photonic crystal structures	[85, 107, 109]
4D Printing	Depends on the 3D printing technology used	Depends on the 3D printing technology used	Shape/function programmable over time; trigger-free self-deformation; imparts dynamic adaptability	Complex design; precise response control required	Realizes adaptive sensors and self-curling electrodes; enhances implant conformability; multi-stimuli responsiveness	[87, 110]
Wet spinning	Fiber diameter: micron-scale	High (continuous fabrication)	Continuous production, controllable diameter, scalable; shear flow induces molecular chain alignment	Limited to fiber geometries; requires solvent treatment and coagulation bath	Realizes axially uniform continuous fiber sensors; post-treatment enhances anisotropy	[88, 90]
Microfluidic spinning	Fiber diameter: micron-scale	Relatively high (continuous fabrication)	Complex morphologies (core-shell, multi-channel); can encapsulate difficult-to-spin polymers; precise laminar flow control	Low throughput; microchannel clogging; requires precise fluid control, challenges in scaling up production; high equipment requirements	Realizes multi-level structured multifunctional fiber sensors; enhances stretchability	[91, 92, 111]
Electrospinning	Down to the nano-scale	Medium (continuous fabrication)	High specific surface area, high porosity; suitable for breathable electrodes	Pure hydrogel spinning difficult; random fiber orientation	Realizes high-sensitivity sensors; excellent breathability; high surface area enhances response	[67, 112]

2.5. Challenges in the fabrication techniques of hydrogels

2.5.1. Trade-off between processing precision and manufacturing throughput. An inherent conflict exists between microscale structural control and scalable production efficiency. High-resolution features like submicron patterns and complex 3D architectures are critical for sensitivity but typically rely on high-precision techniques that increase cost and reduce throughput. Photonic patterning (e.g., photolithography, DLP) enables precise crosslinking but requires strict control and multi-step processing, limiting efficiency^[107]. Conversely, high-throughput methods such as casting, spin coating, and injection molding offer scalability but are constrained by rheology and diffusion to resolutions of tens or hundreds of micrometers; molding also involves costly tooling and potential demolding damage^[94]. This conflict reflects a systemic challenge balancing material rheology, structural control, equipment complexity, and cost. Achieving structural accuracy alongside high efficiency, low cost, and scalability remains a major hurdle for practical sensor manufacturing.

2.5.2. Trade-off between structural complexity and mechanical properties. The pursuit of structural complexity—such as high-aspect-ratio microarrays, porous networks, or oriented fibers—enhances sensitivity and deformability but often compromises polymer network integrity and stress transfer, reducing mechanical robustness. High-aspect-ratio structures risk stress concentration and collapse; porous lattices weaken connectivity and strength; and oriented fibers induce excessive anisotropy, limiting adaptability. Maintaining mechanical reliability while increasing complexity remains a key challenge. Several mitigation strategies have been proposed, including energy-dissipation networks^[113], nanofiller or phase-separation reinforcement^[114], bioinspired structures for crack deflection^[115], and post-treatment to regulate anisotropy and interfacial bonding^[116–119]. These approaches improve local performance but may introduce interfacial incompatibility or long-term instability. The development of multiscale synergistic designs that integrate structural complexity with intrinsic material reinforcement remains essential for achieving both functional performance and mechanical reliability.

2.5.3. Trade-off between precision manufacturing and multiscale integration.

The trade-off originates from the conflict between constructing multiscale functional structures and maintaining overall system stability. Integrating micro/nanoscale functional layers with macroscopic flexible substrates often leads to interfacial stress, adhesion failure, and structural discontinuity due to material mismatches. High-modulus layers may increase signal accuracy but risk delamination from soft substrates, whereas prioritizing flexibility can compromise microstructural integrity. Multi-material fabrication inconsistencies, including mismatches in curing, swelling, and thermal expansion, further induce residual

stress and distortion. Deviations at one scale may also propagate through hierarchical processing. Future efforts should therefore focus on gradient structures, interfacial reinforcement, and coordinated cross-scale manufacturing to achieve both microscale precision and macroscale stability in hydrogel sensors.

3. Strategies for designing hydrogel properties

While traditional hydrogels successfully mimic biological soft tissues in terms of water content and mechanical modulus, they remain intrinsically passive materials composed of inert polymer networks. The inherent electrical insulation, susceptibility to environmental fluctuations (including freezing, dehydration, and swelling), irreversible mechanical damage, and chemical inertness at bio-interfaces constitute the core bottlenecks restricting their application in next-generation flexible electronic devices. The essence of functional design lies in overcoming these intrinsic limitations through molecular engineering and microstructure regulation, thereby endowing the materials with active sensing capabilities and environmental adaptability. This section systematically analyzes the structural design logic of high-performance sensing hydrogels from four critical dimensions: the construction strategies for conductive networks, adaptability design for extreme environments, self-healing strategies, and biocompatibility and antibacterial properties.

3.1. Construction strategies for conductive networks

Imparting excellent charge transport capabilities to insulating polymer networks is a prerequisite for realizing sensing functionalities. Current strategies primarily face a core trade-off: how to introduce highly conductive phases without compromising the intrinsic flexibility and high water content of hydrogels. Depending on the charge carrier type, this field has evolved into two distinct technical routes: electronic conduction and ionic conduction. These approaches encompass diverse functional units, ranging from traditional conductive polymers (CPs) and metal- and carbon-based fillers to emerging materials such as MXenes and ion-based fillers (Figure 5(a)).

Electron-conductive hydrogels: Electron-conductive hydrogels rely primarily on conductive fillers to form a physical percolation network within the polymer matrix. Their performance is governed by the intrinsic conductivity of the fillers, the dispersion state of these fillers, and the interfacial interactions with the matrix. CPs, represented by polyaniline (PANI)^[120–122], polypyrrole (PPy)^[123, 124], and PEDOT:PSS^[125, 126], achieve charge transport through the effective delocalization of conjugated π -electron structures along the polymer backbone^[127]. Compared with rigid inorganic fillers, CPs offer superior organic compatibility, flexibility, and ease of processing. However, hydrophobic conductive chains are prone to macroscopic phase separation

within hydrophilic hydrogel matrices, resulting in discontinuous conductive networks. To address this, common strategies involve directly crosslinking CPs into 3D networks or employing in-situ polymerization to form interpenetrating networks. As shown in Figure 5(b), an ingenious in-situ interpenetrating network strategy was proposed, leveraging controlled micro-phase separation between PVA and PEDOT:PSS^[128]. By inducing partial protonation of polystyrene sulfonate (PSS) via acid treatment to trigger conductive phase precipitation, a one-step self-assembly of continuous nanofiber pathways was successfully achieved. This molecular-level design not only yielded a high conductivity of $906 \text{ mS} \cdot \text{cm}^{-1}$ but also resolved the issue of conductive path fracture common in traditional blend systems under stretching. Liquid metals (LMs)^[129, 130], nanoparticles (NPs)^[131, 132], and nanowires (NWs)^[133–135] are commonly employed to construct efficient charge transport networks within hydrogel matrices. However, the modulus mismatch between rigid metallic fillers and soft matrices often induces severe hysteresis effects. The PAL hydrogel ingeniously leveraged the synergy between LM fluidity and silver nanowires (AgNWs) (Figure 5(c))^[133]. During stretching, the Poisson effect induces lateral contraction of the PVA matrix, forcing anisotropic alignment of the fillers. This structural design not only alleviates the interfacial stress concentration but also achieves a leapfrog increase in conductivity from $4.05 \times 10^{-5} \text{ S} \cdot \text{cm}^{-1}$ to $0.24 \text{ S} \cdot \text{cm}^{-1}$ under strain. On the other hand, carbon-based nanomaterials such as graphene^[136, 137], graphene oxide (GO)^[138, 139], CNTs^[140–142], carbon dots (CDs), and carbon fibers (CNFs)^[143, 144] have become another major class of important fillers due to their excellent environmental stability and biocompatibility. In response to the agglomeration bottleneck of carbon materials in aqueous solutions, sodium lignosulfonate-tannic acid nanospheres (LGTA) were designed via self-assembly^[136]. These nanospheres effectively aided the graphene dispersion and prevented agglomeration, successfully boosting hydrogel conductivity to $0.28 \text{ S} \cdot \text{cm}^{-1}$ while enhancing the mechanical properties (Figure 5(d)). Beyond these classic materials, two-dimensional transition metal carbides/nitrides (TMCNs), represented by MXene and semiconductor materials, are emerging as promising candidates^[145, 146]. MXene nanosheets combine hydrophilic surfaces, large specific surface areas, and metallic conductivity, making them ideal candidates for use in epidermal sensors. Based on these characteristics, researchers integrated nanosheets decorated with antibacterial AgNPs into a guar gum and SA matrix to construct a synergistic AgNP/MXene conductive network (Figure 5(e))^[147]. This design utilized the high specific surface area of MXenes to construct a sensitive layer while employing silver nanoparticles to reduce contact resistance and impart antibacterial properties. Consequently, it achieved highly sensitive capture of both macroscopic joint movements and microscopic ECG/electromyography (EMG) signals.

Ion-conductive hydrogels: unlike electronic conduction, ion-conductive hydrogels utilize free ion migration for charge transport. Their soft and wet characteristics closely mimic those of biological tissues, fundamentally eliminating interfacial impedance mismatch and making them ideal candidates for human-machine interfaces. The mainstream strategy involves dissolving conductive metal salts (e.g., NaCl^[148, 149], LiCl^[150, 151]) or acids (e.g., hydrochloric acid (HCl), sulfuric acid (H₂SO₄)^[152]) into the hydrogel. However, high concentrations of salt solutions often disrupt the hydration layer of polymers, triggering irreversible salting-out effects that lead to material opacification and hardening. To introduce more stable conductive media, ionic liquids (ILs)^[153, 154] have been widely adopted because of their non-volatility and wide electrochemical windows. A notable advancement involves the construction of continuous and independent IL nanochannels within an acrylate matrix via photopolymerization-induced microphase separation (Figure 5(f))^[81]. This phase-separated structure not only achieved an ultra-high ionic conductivity of 0.0323 S·cm⁻¹ but also maintained excellent optical transparency and extremely low mechanical hysteresis (only 0.4% at 50% strain) through precise control of the nanoscale domains. Future design trends are expected to lean towards such structured ion channel designs. This approach aims to balance electrical performance with optical and mechanical properties while leveraging the wide liquid range of ILs to endow materials with additional environmental adaptability.

Electron-ion conductive hydrogels: within electron-ion conductive hydrogels, electronic and ionic conductivities are not isolated phenomena but rather form synergistic transport mechanisms that collectively contribute to the overall conductive performance. For example, in homogeneous conductive hydrogels based on supramolecular host-guest chemistry and self-doped CPs (S-PEDOT), whose sulfonic acid side chains simultaneously served as ion transport sites and electron conduction pathways. This synergistic design significantly increased the conductivity to 5.8 S·cm⁻¹, far surpassing that of conventional PEDOT:PSS hydrogels (0.52 S·cm⁻¹)^[155]. However, in practical applications, the dominant conduction mechanism depends on the material's microstructure, composition, and external environment. Electron conduction typically prevails when highly ordered, continuous conductive networks form. A recent study designed a hydrogel featuring a hierarchical network with an inhomogeneous topological structure. By constructing a composite structure comprising a PEDOT:PSS polycrystalline phase and a hydrated PSS network, the electron hopping mechanism between PEDOT microcrystals was utilized to achieve electron-dominated charge transport. The charge transport rate reached 2×10^6 m·s⁻¹, far exceeding the ion transport rate^[156]. This electron-dominated mechanism offers significant advantages for high-frequency signal transmission and bioelectrochemical

sensing. Conversely, ionic conduction prevails in hydrogels lacking electron percolation networks, such as starch-PVA/GO matrices. While GO alone yields a conductivity of only $7 \times 10^{-4} \text{ S}\cdot\text{cm}^{-1}$, the introduction of imidazole-based ILs increases the value to $0.0198 \text{ S}\cdot\text{cm}^{-1}$ [157]. The result confirms that mobile ions from the IL serve as the primary conductive source, as the electronic contribution of GO remains limited. Therefore, by precisely designing the composition and microstructure of hydrogels, the synergistic and dominant relationship between electronic and ionic conduction can be regulated to meet diverse application requirements, ranging from high-speed electronic sensing to highly sensitive ionic responses.

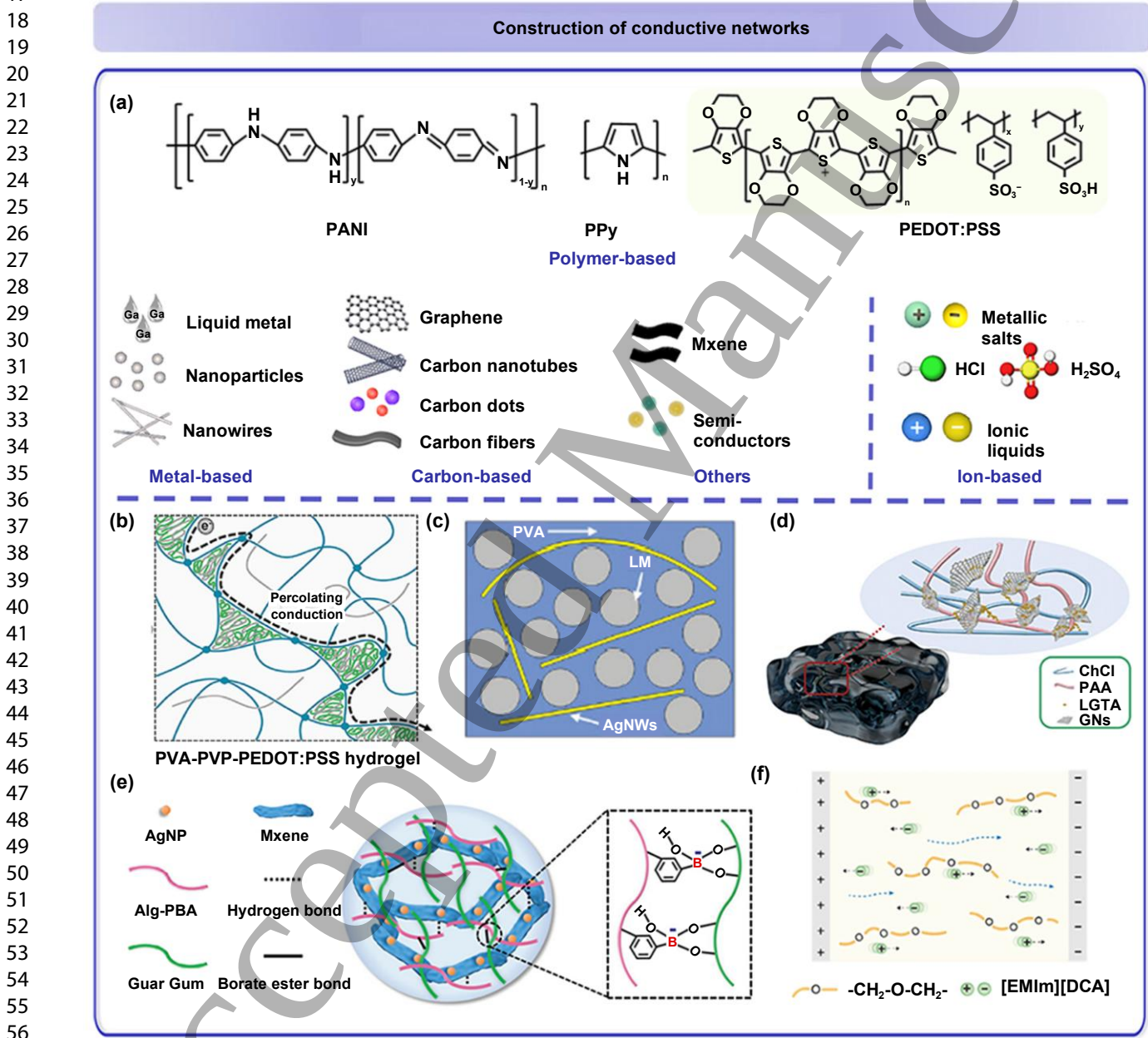


Figure 5. Conductivity properties of the hydrogels. (a) Different types of conductive hydrogels: polymer-based, metal-based, ion-based, carbon-based, and other-type. (b) Interpenetrating conductive polymer network of PEDOT:PSS in the PVA-PVP hydrogel.

[128] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (c) Anisotropic alignment of LM and AgNWs via the Poisson effect for enhanced conductivity. [133] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (d) LGTA nanosphere-assisted graphene dispersion for enhanced conductivity. [136] John Wiley & Sons. © 2025 Wiley-VCH GmbH. (e) AgNPs/MXene conductive network for reduced contact resistance. [147] John Wiley & Sons. © 2022 Wiley-VCH GmbH. (f) Photopolymerization-induced microphase separation for continuous IL nanochannels. Reproduced with permission from [81]. CC BY-NC-ND 4.0.

3.2. Adaptability design for extreme environments

The presence of water in hydrogels is a double-edged sword. While it maintains the material's flexibility and conductivity, it also leads to freezing at low temperatures, hardening due to water loss in dry environments, and excessive swelling underwater. Therefore, structural design at the molecular level to disrupt the hydrogen bond network of water or limit its permeation is key to ensuring stable device operation under all weather conditions.

Anti-freezing and water-retention strategies: addressing freezing and evaporation primarily involves thermodynamically disrupting the tetrahedral hydrogen bond network of free water or lowering its vapor pressure. In hydrogels, water primarily exists as thermodynamically unstable “free water”, which freezes readily at low temperatures and evaporates rapidly in dry environments due to high vapor pressure^[158, 159]. These phenomena lead to material hardening and the loss of electrical conductivity. Consequently, the design of anti-freezing and water-retention hydrogels relies on the introduction of electrolytes with strong hydration capacity (inorganic salts^[160, 161], zwitterions^[162, 163]), strong hydrogen-bond donor/acceptor solvents (organic solvents^[164, 165], acidic substances^[166, 167], deep eutectic solvents (DESs)^[168]), bulky asymmetric ionic solvents (ILs)^[154] or specific biomacromolecules^[169]. These components utilize ion–dipole interactions, hydrogen-bond competition, steric hindrance, or interfacial adsorption to increase the bound water ratio and lower the chemical potential of the solvent system. Such interventions inhibit the ordered arrangement of water molecules and ice nucleation while effectively suppressing moisture loss^[170], ensuring signal stability in the cold or arid environments of hydrogel-based sensors.

Utilizing the colligative properties generated by high concentrations of inorganic salts is a classic and cost-effective strategy. Various metal ions, including Li^+ , Na^+ , Zn^{2+} , Ca^{2+} , and Fe^{3+} , have proven effective^[160, 161, 171, 172]. In a representative study, the synergy between LiCl and a polymer network was exploited to develop PMS hydrogels. The strong hydration of Cl^- and Li^+ disrupted the hydrogen bonds of free water, enabling the material to remain flexible at ultralow temperatures of $-80\text{ }^\circ\text{C}$. Differential scanning calorimetry confirmed

the significant freezing point depression. Furthermore, the strong hygroscopicity of the salt endowed the material with self-regeneration capabilities under ambient humidity (Figure 6(a))^[160]. Although the inorganic salt strategy is efficient, the risk of high-concentration ion crystallization and potential corrosion of metal electrodes limits its long-term application in electronic devices. Inspired by cold-tolerant plants in nature, zwitterions (e.g., betaine and proline) are widely studied because of their unique electrostatically induced hydration effects^[162, 163]. Molecular dynamics simulations confirmed that the highly electronegative oxygen atoms in proline form high-binding-energy complexes with water molecules^[173]. This interaction effectively disrupts the directional arrangement of water, thereby inhibiting ice crystal nucleation (Figure 6(b)). Although this biomimetic strategy exhibits excellent biocompatibility, it requires relatively high concentrations for better anti-freezing.

To further address volatility, introducing organic solvents to construct binary solvent systems has become a more universal choice. Solvents such as Gly, ethylene glycol (EG), and sorbitol utilize their abundant hydroxyl groups (—OH), while dimethyl sulfoxide (DMSO) uses its polar sulfoxide group (S=O) as a strong hydrogen bond acceptor. These functional groups form strong hydrogen bonds with water, significantly reducing the proportion of free water in the system^[164, 165, 174–176]. As shown in Figure 6(c)), in binary-solvent organohydrogel systems, free water molecules unbound to the polymer network undergo rapid dynamic exchange with sorbitol molecules^[174]. Through hydrogen bonding between sorbitol and water, the prepared organohydrogel exhibits superior anti-freezing performance, moisture retention, and responsiveness to environmental humidity. Beyond organic solvents, certain acids also impart wide-temperature tolerance to hydrogels. For instance, phytic acid (PA) and inorganic acids (phosphoric acid (H_3PO_4), H_2SO_4) form robust hydrogen-bonded complexes with water. These interactions increase the ratio of bound water, thereby lowering the freezing point and effectively inhibiting the nucleation of ice crystals^[166, 167, 177]. A H_3PO_4 -mediated CS-PAM hydrogel was developed that exhibited stable operation across a broad range from $-60\text{ }^\circ\text{C}$ to $100\text{ }^\circ\text{C}$, achieving a high ionic conductivity of $0.221\text{ }99\text{ S}\cdot\text{cm}^{-1}$ at high temperatures (Figure 6(d))^[167]. Building on this strategy, H_2SO_4 demonstrates an even more pronounced effect, suppressing freezing and evaporation through a dual-locking mechanism. Its strong binding effect results in the formation of stable hydrates with water (binding energy $>60\text{ kcal}\cdot\text{mol}^{-1}$ for hexahydrate), significantly increasing the water phase-change energy barrier to inhibit ice nucleation and delay evaporation. Concurrently, its spatial confinement effect is achieved by H_2SO_4 reacting with polymer chains (e.g., alginate) to form sulfate ester groups and ionic bonds, anchoring hydrates onto the network and restricting water migration. This synergy enables an ultra-

wide operational range from $-115\text{ }^{\circ}\text{C}$ to $143\text{ }^{\circ}\text{C}$ ^[177]. Despite the effectiveness of these strategies in resolving antifreeze and water retention issues, such solvent systems are not without drawbacks. The potential leakage of small-molecule solvents during long-term service may trigger tissue inflammation, necessitating interface sealing techniques for biomedical applications.

Emerging solvent systems, such as DESs and ILs, offer more thorough solutions. The formation mechanism of DESs primarily relies on extensive hydrogen bond networks, Lewis acid–base interactions, and van der Waals forces between hydrogen bond donors (HBDs) and acceptors (HBAs)^[168]. These strong intermolecular interactions significantly lower the lattice energy of the system, resulting in a eutectic point far below that of the individual components^[178, 179]. Based on this principle, a DES was synthesized using choline chloride (ChCl) as the acceptor and Gly as the donor at a 1 : 2 molar ratio. This formulation markedly extends the liquid working temperature range (-20 – $50\text{ }^{\circ}\text{C}$) and significantly reduces the solvent vapor pressure, thereby imparting exceptional ambient stability to the resulting hydrogel (Figure 6(e))^[180]. In contrast, ILs leverage the steric hindrance between bulky asymmetric organic cations and anions to effectively inhibit regular ionic packing at low temperatures, leading to depressed freezing points and negligible vapor pressure^[154]. This approach has been validated through the construction of a binary system comprising [EMIm]Cl and water, maintaining flexibility and an ionic conductivity of $9.2 \times 10^{-4}\text{ S}\cdot\text{cm}^{-1}$ at $-20\text{ }^{\circ}\text{C}$. The resulting dual-responsive sensor demonstrated high strain sensitivity ($\text{GF} = 2.15$) and temperature detection capability ($-1.845\%\cdot^{\circ}\text{C}^{-1}$)^[181]. However, such devices still confront the risk of signal distortion caused by the crosstalk between strain and temperature in practical applications. Additionally, the high raw material cost of ILs limits their large-scale commercial deployment. Future design efforts should be directed toward developing signal decoupling mechanisms and optimizing low-cost formulations, aiming to simultaneously overcome the bottlenecks in signal fidelity under complex environments and economic feasibility.

Beyond these physicochemical strategies, introducing biomacromolecules such as antifreeze proteins (AFPs) or ice-structuring proteins (ISPs) can inhibit ice growth through surface adsorption^[169, 182]. Such a methodology has demonstrated the capability to maintain hydrogel softness at temperatures as low as $-15\text{ }^{\circ}\text{C}$ ^[183]. Furthermore, chemical modification of the polymer network is an effective approach^[184]. For example, covalently bonding EG through isocyanate groups establishes chemically anchored structures that significantly increase the freeze resistance^[185]. Overall, future designs for anti-freezing and water retention will move beyond single mechanisms toward composite strategies involving multi-solvent synergy and structural encapsulation, aiming to satisfy both electrical stability and biosafety in extreme environments^{[186–}

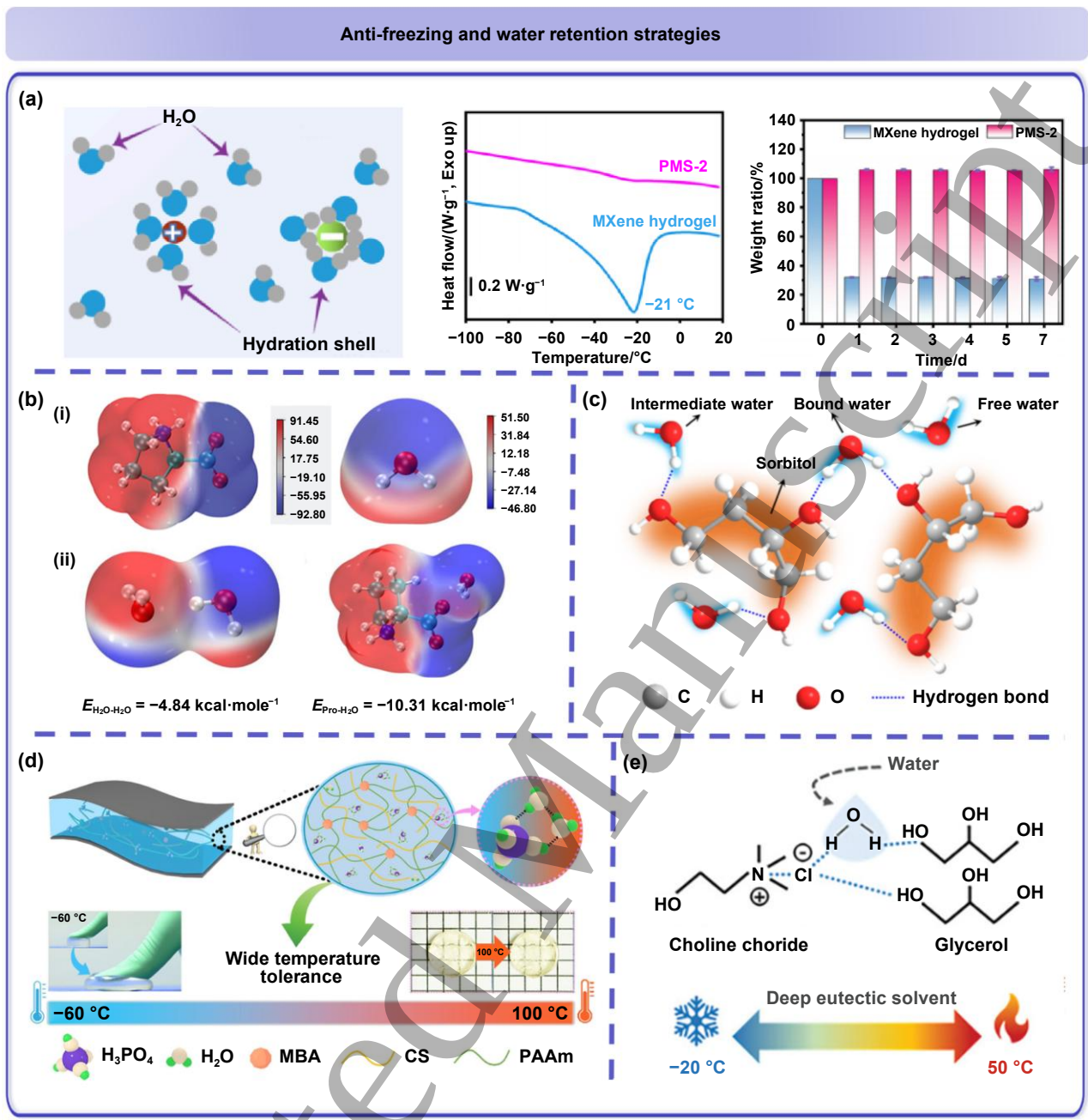


Figure 6. The anti-freezing and water-retention properties of the hydrogels. (a) Introduction of LiCl imparts anti-freezing and self-regenerating properties to PMS hydrogels. Reprinted with permission from [160]. Copyright (2024) American Chemical Society. (b) Molecular electrostatic potential energy surface of Proline, H₂O, H₂O-H₂O, and Pro-H₂O. [173] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (c) Schematic representation of hygroscopic sorbitol molecules adsorbing water molecules by forming hydrogen bonds with them. Reprinted with permission from [174]. Copyright (2022) American Chemical Society. (d) H₃PO₄-mediated CS-PAAm hydrogel for wide-temperature tolerance and high ionic conductivity. Reprinted from [167], Copyright (2021), with permission from Elsevier. (e) DES (choline chloride-glycerol) for hydrogen bond disruption and vapor pressure reduction. [180] John Wiley & Sons.

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Anti-swelling strategies: in underwater or high-humidity application scenarios, the unrestricted swelling of hydrophilic polymer networks leads to increased free volume, diluted ion concentration, and reduced mechanical strength. This consequently causes severe signal drift and structural failure. Therefore, constructing an inherently anti-swelling hydrogel system is essential for maintaining performance stability in such wet environments. Unlike device-level strategies that rely on external physical encapsulation (discussed in Section 4.3), material-level anti-swelling designs introduce two molecular-scale constraint mechanisms to counteract osmotic pressure-driven swelling. On the one hand, reducing the network's affinity for water by adjusting the hydrophilic–hydrophobic balance (e.g., via solvent exchange or hydrophobic modification) drives a transition from a hydrophilic swollen state to a hydrophobic compact state, thereby thermodynamically suppressing swelling. On the other hand, the construction of crosslinked structures with high elastic recovery by increasing the crosslink density effectively resists swelling induced by chemical potential gradients. These strategies limit water molecule migration across interfaces, thereby fundamentally ensuring the stability of hydrogel-based sensors in aqueous environments. The following details the solvent exchange, hydrophobic modification, and high crosslink density strategies involved.

The solvent exchange strategy utilizes the polarity difference between good and poor solvents to modulate polymer chain conformation^[108]. When a gel is transferred from a good solvent to a poor one, chain–chain interactions dominate over chain–solvent interactions. This drives the molecular chains to undergo microphase separation and collapse into a dense physically cross-linked network. This process shifts the gel from a hydrophilic swollen state to a hydrophobic compact state, thereby thermodynamically suppressing swelling. Specifically, a DMSO-to-water solvent exchange strategy was implemented to induce controlled aggregation of the hydrophobic segments, yielding a swelling ratio of 4.79% (Figure 7(a))^[61]. Furthermore, such an approach facilitates the fabrication of high-strength exogels, with the primary advantage being the attainment of exceptional mechanical strength^[189]. However, excessive network densification may compromise stretchability and ion transport rates. Therefore, precise control over the degree of exchange is essential.

Hydrophobic modification strategy offers a direct route for constructing endogenous barriers by leveraging the entropy-driven aggregation of hydrophobic groups^[190]. The introduced hydrophobic groups spontaneously aggregate to minimize contact with water molecules, thereby significantly reducing the material's affinity for water and enhancing its inherent hydrophobicity. For instance, high-density hydrophobic

microdomains were successfully incorporated within hydrogels through the ultrasonic-assisted introduction of hydrophobic monomers^[191]. This approach maintained the swelling ratio at a low level of 26% and endowed the material with excellent underwater adhesion due to the outward orientation of hydrophobic groups (Figure 7(b)). Simultaneously, the preparation of gels from hydrophobic monomers with poor water solubility typically requires strategies involving organic solvents, ILs, or hydrophobic DESs. In one representative example, a multifunctional PEA-AA-MAm/IL gel was synthesized via one-step UV-initiated copolymerization of 2-phenoxyethyl acrylate (PEA), acrylic acid (AA), and methacrylamide (MAm). The resulting hydrogel exhibited exceptional resistance to swelling and maintained long-term stability under underwater conditions^[192]. However, hydrophobic modification requires a delicate balance between anti-swelling performance and ionic conductivity. Excessive hydrophobic content may lead to the blockage of ion channels. Therefore, constructing a bicontinuous phase structure is a key direction for future design. Such structures should comprise hydrophilic channels for ion transport and a hydrophobic backbone for swelling resistance. These internal regulation strategies circumvent the interfacial failure risks often associated with multilayer encapsulation structures.

A high cross-linking density strategy is an effective strategy to improve the anti-swelling performance of hydrogels. Multiple cross-linked networks are constructed via dynamic non-covalent interactions (hydrophobic interactions, electrostatic interactions, metal coordination, and so forth), and the Hofmeister effect not only increases the cross-linking density but also endows hydrogels with superior anti-swelling properties and mechanical stability^[193, 194]. For example, the formation of nanocrystalline domains in PVA chains was promoted via the Hofmeister effect by employing H₂SO₄ and poly(sulfobetaine methacrylate) (PSBMA). These nanocrystalline domains served as physical cross-linking points and significantly improved the network's resistance to deformation. Consequently, the sensor maintained a stable performance after 30 days of immersion in seawater (Figure 7(c))^[152]. Similarly, an anti-swelling, high-strength hydrogel (PAM/SA/Zr⁴⁺/CDs, PSZC) was developed by integrating mechanical stretching, Zr⁴⁺ coordination, and carbon dot doping strategies. This approach created a dense, oriented network that effectively suppressed water permeation, significantly enhancing the anti-swelling properties of the hydrogel^[195]. Notably, while increasing the crosslinking density improves anti-swelling performance, it may also reduce flexibility, increase brittleness, decrease stretchability, and weaken interfacial adhesion. Therefore, a key challenge for practical applications of hydrogel-based flexible sensors is to balance environmental stability with core properties such as flexibility, stretchability, adhesion, and biocompatibility.

Environmental tolerance strategies, including anti-freezing, water retention, and anti-swelling, significantly dictate the performance and application scope of hydrogels (Table 2). While modulating polymer networks or introducing salts and zwitterions broadens operating temperatures and enhances moisturizing capacity, the resulting high hydrophilicity often compromises anti-swelling stability. Furthermore, high salt concentrations may trigger secondary issues like precipitation or corrosion. In contrast, organic solvent-based and acid-based hydrogels offer superior low-temperature tolerance and water retention, although underwater functionality necessitates the integration of hydrophobic microdomains or highly crosslinked networks. Organohydrogels prepared via solvent exchange achieve a synergistic balance of these properties, whereas biomacromolecular agents provide excellent biocompatibility despite their limited long-term stability. DES have emerged as cost-effective, non-toxic alternatives to ILs, bypassing the toxicity and leakage risks associated with the latter. Ultimately, reinforcing anti-swelling through hydrophobic modification or dense crosslinking requires balancing environmental tolerance with mechanical and conductive performance. Given that single strategies rarely satisfy all the requirements, integrated multi-mechanism designs have become the primary pathway for multi-environmental adaptability. For instance, a P(SBMA-*co*-AA) hydrogel matrix combining DES (BMIMCl/solketal) with a high crosslinking density and CTAB units achieved a freezing point of $-42\text{ }^{\circ}\text{C}$ and retained 81.3% of its shape after 15 days. It exhibited a swelling ratio of only $\sim 3\%$ after 30 days in water, with 91.8% mechanical retention^[194]. Similarly, an IL-based hydrogel utilizing hydrophobic modification, such as P([TFEA]-*co*-[AAHA])[TFSI]) and [EMIM][TFSI] combination, operated stably from $-40\text{ }^{\circ}\text{C}$ to $40\text{ }^{\circ}\text{C}$ with a swelling ratio of $\sim 1.2\%$ after 10 days^[196]. Furthermore, integrating ILs with Al^{3+} coordination in a P(AM-*co*-VBIMBr-*co*-IA) hydrogel network resulted in more than 70% water retention after 15 days and a swelling ratio of $\sim 4.5\%$ after 30 days^[197]. These cases demonstrate that the merging anti-freezing solvents, hydrophobic domains, and coordination chemistry enable the construction of all-weather hydrogel sensors for diverse applications ranging from extreme environmental monitoring to biomedicine.

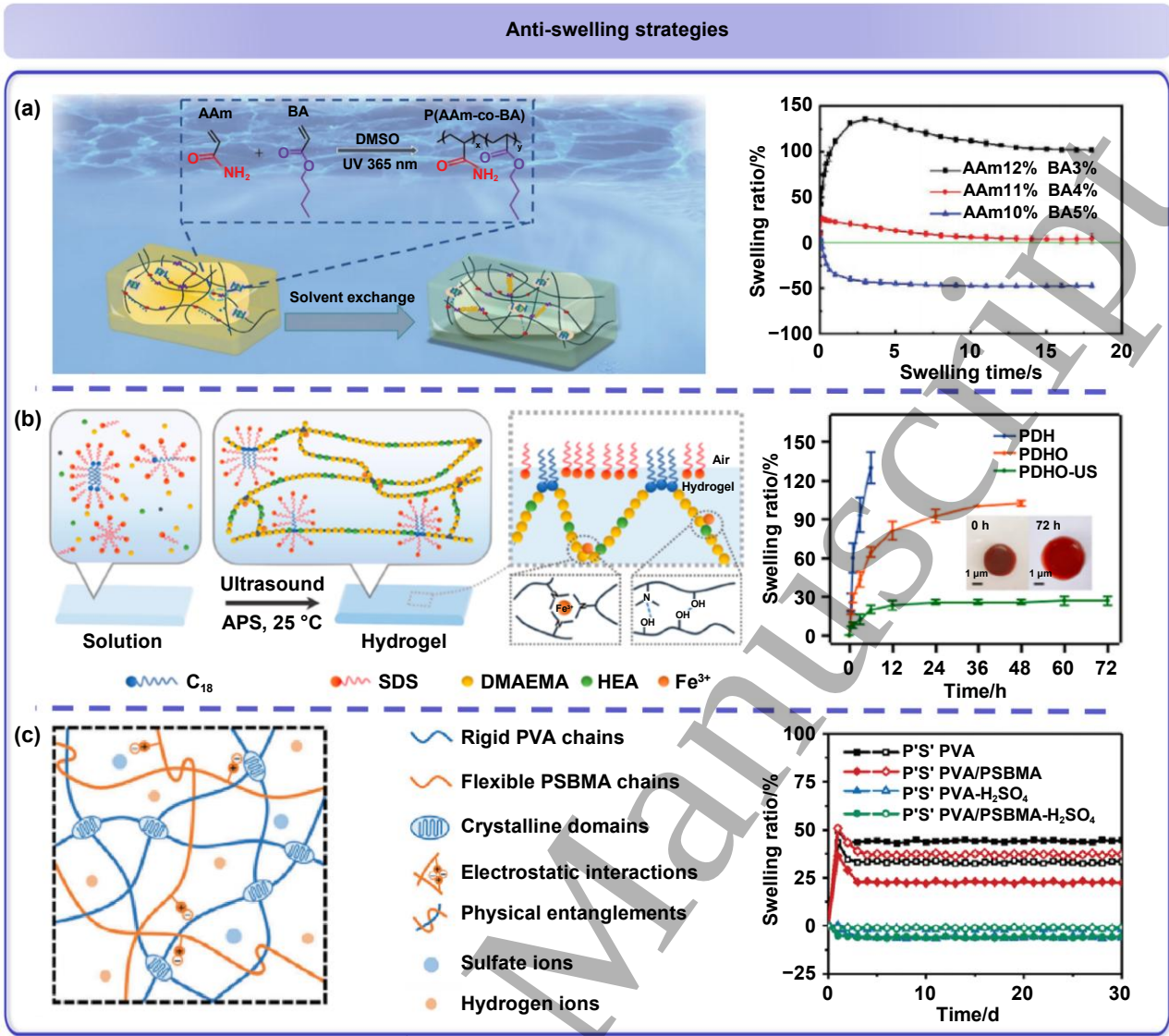


Figure 7. The anti-swelling property of hydrogels. (a) DMSO-to-water solvent exchange process and the resulting swelling curve of the gel. ^[61] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (b) Ultrasonic-assisted hydrophobic monomer incorporation for swelling suppression and the resulting swelling curves of hydrogels. Reprinted with permission from^[191]. Copyright (2022) American Chemical Society. (c) Hofmeister effect-induced nanocrystalline domains for enhanced cross-linking density and anti-swelling performance. ^[152] John Wiley & Sons. © 2024 Wiley-VCH GmbH.

Table 2. Comparative analysis of representative strategies for hydrogels with extreme environmental adaptability.

Strategy category	Typical hydrogel compositions	Environmental tolerance	Key advantages	Primary limitations	Ref.
Inorganic salt-enhanced	PAM/CTAB/CMC/ Super p/LiCl	Freezing point: -110°C ; maintaining excellent stretchability (-90°C)	Remarkable anti-freezing efficacy; low cost; simple preparation; biocompatibility	High salt concentrations prone to crystallization; potential metal electrode corrosion; limited long-term water-retention; poor anti-swelling efficacy	[198]
	PAM/NVP/AGE/CaCl ₂	Not completely frozen (-60°C); $\sim 60\%$ (>90 h at 30°C and 50% RH) $\sim 125\%$ (40 h in water)			[199]
	PAM/Zn(Ac) ₂ /KAc	Operating range: -20 – 100°C ; $\sim 70\%$ (60 min at 60°C)			[200]
Zwitterion-based	PVA/L-proline/HA/ CNCs@BSA	Freezing point: -48.3°C ; $\sim 86.4\%$ of original stretchability (-30°C); $\sim 73.8\%$ (7 days at 25°C); $\sim 36.5\%$ (450 min in water)	Bio-inspired design; simple preparation; remarkable anti-freezing efficacy; excellent biocompatibility; non-cytotoxic	Requires high concentrations; limited anti-swelling efficacy	[201]
	PAM/PVA/betaine/As	Freezing point: -68°C ; $\sim 88\%$ (15 days) Operating range: -70 – 60°C ;			[202]
Organic solvent-based	PVA/AP/LS/Gly	$\sim 97.7\%$ of original stretchability (-20°C) $\sim 88.5\%$ of original stretchability (60°C)	Simple preparation; remarkable anti-freezing efficacy; excellent long-term water retention	Requires composite anti-swelling design; potential leakage of small molecule solvents; certain chemical inertness impairs conductivity; some raise potential safety concerns	[203]
	P(AM-co-AA)/PANI/Gly	Freezing point: $<-100^{\circ}\text{C}$; Operating range: -18 – 60°C ; $\sim 97.3\%$ (7 days at 25°C and 60% RH) $\sim 59.1\%$ (6 h at 60°C)			[204]
	PAM/CS/H ₃ PO ₄	Operating range: -60 – 100°C ; approaching the original mechanical properties (-60°C , 100°C); $\sim 84\%$ (15 days in air)			[167]
Acid substance-based	PAM/alginate/H ₂ SO ₄	Operating range: -115 – 143°C ; $\sim 54.5\%$ of original stretchability (-100°C); $\sim 72.7\%$ of original stretchability (140°C)	Wide temperature range; high ionic conductivity from proton release ; multifunctional (antibacterial, adhesive)	Requires composite anti-swelling design; strong acids (pH <2) may hydrolyze and degrade polymer networks over time; potential corrosion of electrodes or tissues at low pH	[177]
Deep eutectic solvent-based	DES (ChCl/Gly)/PAM, PA/Trehalose	Freezing point: $<-70^{\circ}\text{C}$; $\sim 94.5\%$ (48 h at 50°C)	Remarkable anti-freezing efficacy; water retention; environmentally friendly; low-cost biocompatibility; low volatility	Hygroscopicity influence; requires composite anti-swelling design; possibility of escape upon contact with water-containing solvents	[180]

Table 2. Continued.

Strategy category	Typical hydrogel compositions	Environmental tolerance	Key advantages	Primary limitations	Ref.
Ionic liquid-based	PVA/Mxene/VBImBr/ IL cross-linker (Ph-3MVIm-Br)	Operating range: -60–80 °C; ~75% of original stretchability (-60 °C); ~135% of original stretchability (80 °C); ~81.2% (30 days in air) Freezing point: -90 °C; approaching the original mechanical properties	Environmental stability; wide temperature range; remarkable anti-freezing efficacy; excellent long-term water retention; partial anti-swelling	Requires composite anti- swelling design; high cost; potential biotoxicity of some ILs; some complex synthesis of some ILs; leakage risk of ILs	[205]
	[Ch][Gly]/PVA/Cellulose	(-20 °C); >85 % (7 days at 37 °C); ~25% (over 28 days in water)			[206]
	PAM-co-MAAc/BMIP/ NaCl	-30 °C (~23.7% of original stretchability) -45 °C (~69.1% of original stretchability); Maintaining unchanged of the original mass and volume (10 days in water)			[207]
Solvent exchange	DMSO → Water P(MMA-co-HEA), DMSO → CaCl ₂ solution P(NASC-co-AA)/ PVDF-TrFE	~1.9% (100 days in water); approaching the original stretchability (30 days in water)	Outstanding anti-swelling; partial anti-freezing	Excessive densification affects flexibility and ion conduction; requires precise control of exchange time	[208]
					[108]
Hydrophobic modification	P(AM-co-HEMA-co-MEA)/ DMSO/CNT-TA-Fe ³⁺	Freezing point: <-60 °C; <2.3 % (15 days in water)	Outstanding anti-swelling; partial anti-freezing; no post-treatment required	Excessive hydrophobicity blocks ion channels, reducing conductivity; requires balance between hydrophobicity and hydrophilicity	[209]
High crosslink density	PAM/SA/CDs/Zr ⁴⁺	~1.7% (30 days in water); ~6.0% (30 days in water); (100 ± 5)% of original tensile strength, stretchability, Young's modulus, and toughness (30 days in water)	Outstanding anti-swelling; significant mechanical stability; can achieve water retention	Requires composite anti- freezing design; potential increase in brittleness	[210]
	PVA/PSBMA/H ₂ SO ₄				[152]

3.3. Self-healing strategies

During the long-term operation of flexible electronic devices, repeated mechanical deformation inevitably induces microcracks in hydrogel networks, leading to conductive pathway disruption and device failure. Endowing hydrogels with self-healing capabilities essentially involves incorporating dynamic bonds that can dissipate energy and undergo reversible reorganization, enabling spontaneous functional recovery through molecular chain diffusion and re-entanglement at damaged interfaces. Current design strategies primarily focus on dynamic covalent bonds, non-covalent interactions, and their synergistic effects.

Self-healing strategies based on dynamic covalent bonds enable hydrogel self-healing through reversible chemical bonding, where covalent bonds can reform after breaking^[211]. Common chemical motifs include imine bonds^[212], acylhydrazone bonds^[213], borate ester bonds^[153], disulfide bonds^[214], and Diels-Alder (DA) reactions^[215] (Figure 8(a)). Imine bonds, formed via reversible condensation between amines and carbonyl compounds (e.g., aldehydes and ketones), exhibit dynamic covalent characteristics. Owing to their reversible equilibrium properties, they are widely utilized to fabricate self-healing hydrogels^[216, 217]. In 2011, a breakthrough study demonstrated rapid gelation of hydrogels based on the Schiff-base reaction and discovered that the dynamic crosslinking mechanism endowed the hydrogels with self-healing properties. This finding subsequently attracted widespread attention^[218]. Acylhydrazone bonds are established through the reversible condensation of hydrazines with aldehydes/ketones. While the formation mechanism is analogous to that of imine bonds, acylhydrazone linkages exhibit enhanced stability^[213]. Consequently, acidic conditions or thermal stimuli are typically required to facilitate the dynamic reversibility of acylhydrazone bonds, thereby enabling self-healing properties within the hydrogel^[219, 220]. In boron-based systems, electron-deficient boron atoms in boronic acid (or its derivatives) form reversible boronic ester bonds with cis-diols^[221, 222]. These dynamic bonds undergo continuous reversible rearrangement near damaged regions, maintaining structural equilibrium and self-healing capacity. For instance, through the integration of dynamic boronic ester bond exchange and physical ball milling, the fabricated PMP hydrogel exhibited nearly 100% mechanical recovery within 120 s at room temperature. These results demonstrate the significant potential of dynamic boronic ester bonds for rapid reconstruction^[223]. Disulfide bonds, as another representative class of dynamic covalent bonds, derive their dynamic characteristics primarily from exchange reactions triggered by thermal, UV, or redox stimuli. Under external stimulation, disulfide bonds at the fracture interface undergo homolysis or thiol–disulfide exchange reactions, which induce dynamic reorganization of the polymer network, thereby achieving self-healing^[224]. Owing to their excellent compatibility with polymer matrices, the integration of disulfide

bonds into hydrogel networks with a low glass transition temperature enables temperature-reversible self-healing. Furthermore, the Diels-Alder (DA) reaction, a classical [4+2] cycloaddition, generates six-membered ring structures through the reaction between a diene and a dienophile. Its inherent dynamic reversibility enables the introduction of self-healing properties into hydrogels. However, the retro-DA reaction typically requires elevated temperatures, which limits its practical applicability in epidermal environments^[225, 226]. Therefore, designing such materials requires carefully balancing dynamic activity with environmental stability to suit complex application scenarios.

Unlike covalent bonds that require specific stimuli, self-healing hydrogels based on non-covalent interactions rely primarily on dynamic reversible physical forces with low activation energies, enabling autonomous healing at room temperature. These non-covalent interactions primarily include hydrogen bonds^[227], metal coordination^[228], ionic interactions^[229], host-guest interaction^[230], and hydrophobic association (Figure 8(b))^[194]. Hydrogen bonds represent the most fundamental mechanism. With a bond energy of $\sim 42 \text{ kJ}\cdot\text{mol}^{-1}$, it lies thermodynamically between van der Waals forces and strong chemical bonds^[231]. The picosecond-scale dissociation and recombination kinetics allow for instantaneous repair. However, hydrogels composed solely of hydrogen bonds suffer from relatively low mechanical strength. Furthermore, the post self-healing properties of such hydrogels are highly susceptible to environmental conditions (such as temperature, humidity, pH, etc.), often resulting in structural failure or fracture. In contrast, metal-coordination bonds provide hydrogels with both structural stability and dynamic reversibility through Lewis acid-base interactions. Specifically, oxygen/nitrogen-containing functional groups (e.g., carboxyl, catechol, and amino groups) within the network act as Lewis bases by donating electron pairs, while metal ions (e.g., Ca^{2+} , Al^{3+} , Fe^{3+} , and Zr^{4+}) serve as Lewis acids by accepting electrons, forming coordination cross-linking points^[232, 233]. This bonding not only significantly enhances the mechanical strength of hydrogels but also enables self-healing of damaged interfaces through reversible dissociation and reconstruction without the need for external stimuli. The PAA- Fe^{3+} hydrogels utilized this mechanism to achieve self-healing^[228]. Ionic interactions construct dynamic cross-linked networks through electrostatic attraction between charged polymers and counter-ions (polymers, ions, and nanomaterials)^[229, 234–237]. The healing mechanism relies on the redistribution of free ions at the fracture interface and the restoration of charge balance. Owing to its higher bond energy than hydrogen bonds, this mechanism can enhance both the mechanical strength and the performance stability of self-healing hydrogels. Furthermore, ionic interactions include ion-dipole^[238] and dipole-dipole interactions^[239]. Ionic crosslinking is simple and efficient, yet it remains susceptible to charge

shielding in high-ionic-strength physiological environments, potentially resulting in network disintegration. To improve healing selectivity, host–guest interactions employ multiple non-covalent interactions, including hydrogen bonding, electrostatic attraction, π -electron stacking, and hydrophobic effects, to establish reversible crosslinking between cyclic host structures (e.g., cyclodextrins and crown ethers) and geometrically complementary guest molecules (e.g., adamantane derivatives and ferrocene compounds)^[240,241]. For example, the β -cyclodextrin (β -CD) was grafted onto silk fibroin and PAA chains, utilizing host-guest inclusion between the hydrophobic cavity and amino acid residues^[230]. This endows the hydrogel with autonomous healing capabilities without external intervention. The “lock-and-key” mechanism prevents non-specific aggregation and ensures structural fidelity during repair. Additionally, the self-healing mechanism of hydrophobically associated hydrogels relies on entropy-driven hydrophobic interactions. Hydrogels self-assemble into micelle crosslinks at the damage interface, reconstructing the network through the rearrangement of hydrophobic segments.

However, single non-covalent interactions often face a trade-off between mechanical strength and healing efficiency, limiting their utility in high-performance devices. Consequently, constructing networks based on synergistic dynamic interactions has become a key breakthrough^[242–246]. For example, hydrophobic interactions, hydrogen bonding, and electrostatic interactions were integrated into a DES/water system via one-step photopolymerization, achieving 94% self-healing within 48 h^[194]. Similarly, a supramolecular network was developed by incorporating Fe^{3+} coordination, imine bonds, and borate ester bonds. Multiple mechanisms not only provided excellent mechanical strength but also ensured its self-healing capability after repeated damage^[247]. These findings highlight the pivotal role of multi-dynamic interactions in hydrogel design. Through further optimization of their synergistic effects, it is expected that more functional and practical self-healing materials can be developed.

3.4. Biocompatibility and antibacterial properties

For flexible sensors attached to the skin or implanted in vivo, bio-interfacial safety is a prerequisite for application. Biocompatibility ensures that materials maintain functionality without eliciting toxicity, inflammation, or immune rejection during interactions with biological systems, while antibacterial properties safeguard application safety through pathogen elimination. These characteristics originate from two intrinsic attributes: (1) the high water content allows it to remain moist in contact with biological tissues without causing dryness or irritation, (2) properties similar to those of the natural extracellular matrix promote cell attachment, proliferation, and differentiation^[248,249]. Hydrogels with biocompatibility are typically composed

of natural or synthetic polymers (such as PAM, polyethylene glycol (PEG), and PVA). Antibacterial hydrogels include inorganic nanoparticle-based hydrogels, including silver nanoparticles (Ag NPs), gold nanoparticles (Au NPs), and zinc oxide nanoparticles (ZnO NPs); hydrogels incorporating organic antibacterial agents, such as PA and tannic acid (TA); and intrinsically antibacterial hydrogels, which encompass natural polymers (e.g., CS), synthetic polymers (e.g., quaternary ammonium compounds, polylysine, and stimuli-responsive polymers), zwitterionic polymers (e.g., polysulfobetaine), and antimicrobial peptides. Flexible electronic devices fabricated from such materials significantly reduce foreign body reactions during prolonged wear/implantation, thus fulfilling the continuous monitoring requirements of intelligent healthcare systems. For instance, a multifunctional conductive organic hydrogel (designated as C-CST/PMC) synthesised using silk nanofibres (SNFs), catechol-conjugated succinylcholine (C-CS), TA, $\text{MgCl}_2/\text{CaCl}_2$, and PVA has enabled long-term wearable motion-sensing applications^[250]. The synergistic effects of C-CS, SNFs, and TA endowed C-CST/PMC with exceptional antibacterial activity, antioxidant capacity, and biocompatibility. These properties were validated via Cell Counting Kit-8 (CCK-8) assays and live/dead staining, as demonstrated in Figure 8(c). Similarly, incorporating PA into PAM/AA hydrogels confers both tissue-mimetic mechanical properties and excellent biocompatibility^[96]. As shown in Figure 8(d), PA-containing hydrogels exhibited significant inhibitory activity against *E. coli* and *S. aureus*. Concurrently, skin irritation tests revealed intact epidermis and no significant inflammatory infiltration in the hydrogel-treated group, confirming the material's outstanding biocompatibility.

Although biocompatible and antibacterial hydrogels demonstrate significant potential for biomedical applications, current research faces critical challenges in balancing long-term antimicrobial efficacy with low cytotoxicity and controllable in vivo degradation. The key limitations include insufficient antimicrobial selectivity, toxic degradation byproducts, and instability under physiological conditions. Furthermore, in applications such as implantable electronic devices, hydrogels must simultaneously maintain stable conductivity and mechanical properties. Future solutions may involve smart stimulus-responsive systems (e.g., pH- or reactive oxygen species (ROS)-triggered release), bioinspired antibacterial structures (e.g., peptide mimics), and hybrid degradable materials to optimize multifunctional performance.

In conclusion, hydrogel-based strategies have emerged as a pivotal approach for tailoring material characteristics. As detailed in Table 3, significant progress in diverse hydrogel compositions has realized functionalities such as conductivity, environmental robustness, self-healing, and biofunctional traits (e.g., biocompatibility and antimicrobial activity). Such advancements significantly extend the utility of hydrogels

in flexible electronics and biomedical engineering, while providing a conceptual roadmap for next-generation smart materials. Driven by the convergence of multi-scale architecture and intelligent control, future hydrogel technology will shift its focus from individual property enhancement to holistic functional integration and self-adaptability, providing essential scaffolding for the realization of sophisticated intelligent interactive material systems.

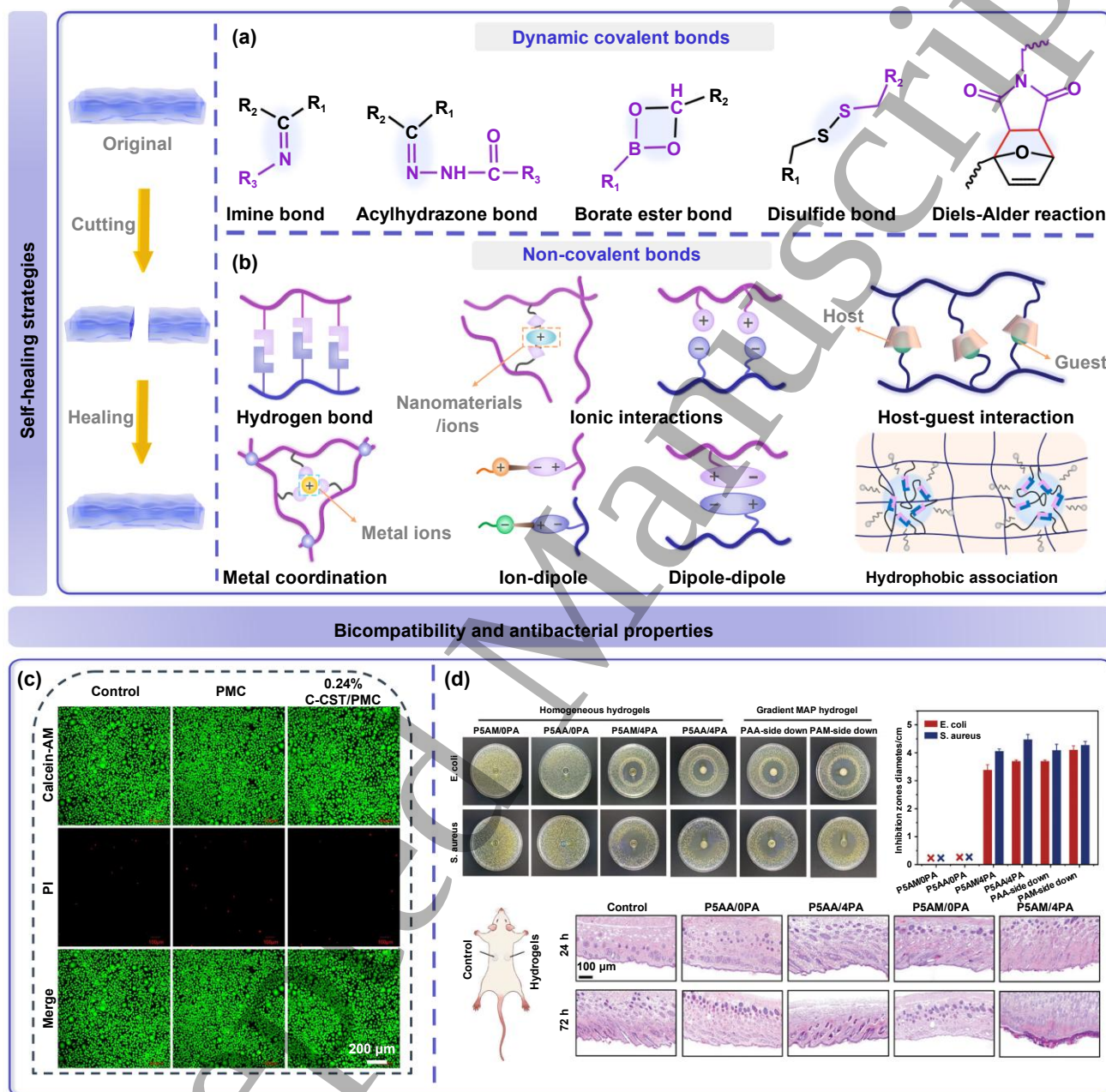


Figure 8. The self-healing, biocompatibility, and antibacterial properties of the functional hydrogel. (a) Self-healing strategies based on dynamic covalent bonds. (b) Self-healing strategies based on non-covalent bonds. (c) The fluorescence images of calcein-AM/PI staining of L-929 cells after 48 hours of incubation with the sample extracts. Reprinted from^[250]. Copyright (2024), with permission from Elsevier. (d) The antibacterial properties and biocompatibility of PA-based hydrogels. ^[96] John Wiley & Sons. © 2025 Wiley-VCH GmbH.

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3.5. Challenges in strategies for designing hydrogel properties

3.5.1. Trade-off among high conductivity, mechanical properties, and biocompatibility. Integrating high conductivity, robust mechanics, and biocompatibility requires a systematic balance among conductive components, network architecture, and biosafety. Electronic conductors (e.g., CPs and graphene) can significantly enhance conductivity; however, filler aggregation, insufficient surface functionalization, or residual impurities may induce cytotoxicity and increase the local modulus, thereby compromising mechanical matching with tissues^[133, 251]. Metallic fillers achieve ultrahigh conductivity (up to $1\ 739\ \text{S}\cdot\text{cm}^{-1}$) but risk metal ion release (e.g., Ag^+)^[134]. LMs exhibit excellent electrical continuity and strain adaptability, with elongations exceeding 2 000%, but they still face challenges such as poor wettability and leakage risk^[252, 253]. Ionic conductors provide softness but high concentrations may disrupt the cellular osmotic balance. The data summarized in Table 3 further reveal the performance distribution of different conductive systems in terms of electrical conductivity and stretchability. Polymer-based hydrogels show the broadest conductivity range (10^{-3} – $10^2\ \text{S}\cdot\text{cm}^{-1}$), with a maximum of $101.4\ \text{S}\cdot\text{cm}^{-1}$. Carbon-based systems typically range from 10^{-5} to $10^{-1}\ \text{S}\cdot\text{cm}^{-1}$. Metal-based systems achieve the highest conductivity but exhibit relatively limited stretchability. In contrast, ionic- and composite-based hydrogels present moderate conductivity while achieving high elongation (>1 500%), indicating superior compliance. Overall, achieving high conductivity often relies on high filler loading or dense conductive networks, which may compromise softness and biosafety. Conversely, prioritizing mechanical and biological compatibility can restrict charge transport efficiency. Therefore, achieving synergistic multifunctionality requires the construction of efficient and stable conductive pathways, combined with network topology regulation, interfacial reinforcement, controlled filler distribution, and ion–electron synergistic transport design, to optimize performance while maintaining biocompatibility and mechanical flexibility.

3.5.2 Trade-offs among environmental tolerance, mechanical properties, and ionic conductivity. This stems from the intrinsic conflict between thermodynamic stability and kinetic transport. While anti-freezing and water-retention agents suppress ice formation and evaporation, they often compromise charge transfer and structural integrity. For instance, excessive inorganic electrolytes trigger salting-out and ion association, which reduce the effective conductivity and induce brittleness or opacity through polymer chain aggregation^[254]. Similarly, high-viscosity co-solvents impede ion migration and dilute the network skeleton, leading to slower response times and mechanical collapse. Overdoping with viscous ILs also causes leaching, compromising shape stability and cycle life. In terms of anti-swelling activity, hydrophobic densification or increased

crosslinking often sacrifices fundamental performance^[205, 255]. High crosslinking density enhances strength but obstructs ion migration, whereas low density facilitates conduction at the expense of structural stability^[256]. Critically, traditional hydrophobic modification causes obstructive microphase separation; the resulting hydrophobic domains act as physical barriers that increase path tortuosity and ionic resistance. These large hydrophobic clusters further reduce the transparency and induce significant hysteresis and residual strain^[257]. Overcoming these antagonisms requires a shift from “functional stacking” to “functional decoupling.” Recent advances demonstrate that inducing bicontinuous phase separation to create dual-continuous ion channels can simultaneously enhance mechanical and electrochemical performance^[62, 258]. This suggests that phase separation is not inherently detrimental and that the key lies in the precise topological control of phase domains. Consequently, the core future challenge is constructing multiscale hierarchical networks. By optimizing hybrid crosslinking modes, network topology, and hydrophilic–hydrophobic balance, researchers should establish a precise equilibrium between segmental mobility, structural stability, and energy dissipation.

3.5.3 Trade-off between multifunctional integration and long-term stability. Multifunctional hydrogels integrate conductive units, reversible crosslinks, antimicrobial agents, and adaptive regulators to enable signal transmission, self-healing, environmental adaptability, and biointerface control. However, long-term use risks component migration, phase separation, or aggregation, disrupting network uniformity. Dynamic reversible bonds support healing but may cause network rearrangement and stress relaxation, while stable covalent bonds enhance durability but limit adaptability and healing^[259]. Conductive pathways and functional additives may suffer interfacial failure under repeated deformation, leading to signal drift^[260]. Therefore, a fundamental trade-off exists between dynamic tunability and long-term stability. Future designs should emphasize multiscale topological engineering, synergistic stable-dynamic networks, and controlled anchoring of functional components for sustained multifunctional performance.

Table 3. Demonstration of the properties of hydrogels with different components.

Hydrogel type	Materials	Conductivity ($S \cdot cm^{-1}$)	Stretchability /%	Anti-freezing and water retention	Anti-swelling	Self-healing	Biocompatibility and antibacterial	Ref.
Polymer-based	PEDOT:PSS/EG	101.4	-	-	-	-	√	[261]
	Gelatin/PVA/PEDOT: PSS/Co-NPC/Gly	0.027 3	>400	>90% (14 days at room temperature)	-	√	Cell viability: 99%	[262]
	BC/PVA/PAA/TA /PEDOT:PSS	11.1	605.8	-	√	-	Cell viability: >94%	[263]
	P(PEG-co-AA)/PANi/Gly	0.074 3	>1 000	-28 °C >80% (3 days at 20 °C)	-	93.3% (10 min)	-	[121]
	ANFs/PPy/PVA	~80	~36	-	-	-	√	[124]
Carbon-based	PHEMA/PVP/PPy	7.8×10^{-3}	451.4	-	21.1% (120 h)	-	-	[256]
	PDA/PAM/GO/EG	0.173	1 100	-20 °C	-	-	-	[139]
	Fe@C/PVA/TA/Starch	0.019 9	240	-60 °C	-	>99% (10 min)	-	[264]
	PDA@CNT/PAM	2×10^{-5}	>700	-	-	97.3% (6 h)	Cell viability: >90%	[140]
	AgNWs/PVA	~958	>300	-	<10% (24 h)	-	-	[134]
Metal-based	PHEA/Ag flake/PVA, PEDOT: PSS/PDA	~571.43	128	-	-	-	√	[265]
	PAA/ACC/LM@TA	2.83×10^{-4}	>2 000	-	-	96.4% (<1 s)	Cell viability: 96.84%	[253]
	PVA/AgNWs	1 739	480	-	-	-	Cell viability: ~98%	[266]
	PAA/Alkali lignin/DL@LM	2.14×10^{-3}	2 315	-	-	96% (12 h)	Cell viability: >90%	[267]
	Cellulose/LiBr/ZnCl ₂	0.096 8	150	-40–80 °C √	-	-	-	[268]
Ion-based	GT ₅ /DACD ₂ /borax/ fusidic acid/PBS	1.2×10^{-4}	67.8	-	-	-	Cell viability: 80% <i>S. aureus</i> : 100%	[269]
	P(AA-co-HEAA-co- MMA)/CS/CTAB/PANa	2.9×10^{-3}	~1 500	64% (15 days)	25% (15 days)	-	-	[270]
	PAA/Fe ³⁺	0.012 2	1 708.7	-	-	88% (1 h)	-	[228]
	P(ATFE-co-NIPAm), [EMIM][TFSI]	2.04×10^{-3}	1 800	-21 °C √	√	Up to 96% (2 h)	-	[271]
	PVA/TA/P(AM-co- VBIMBr-co-IA), Al ³⁺	0.021 65	854	-32.34 °C >70% (15 days at 25 °C and 60 % RH)	-4.5% (30 days)	-	-	[197]

Table 3. Continued.

Hydrogel type	Materials	Conductivity ($\text{S}\cdot\text{cm}^{-1}$)	Stretchability /%	Anti-freezing and water retention	Anti-swelling	Self-healing	Biocompatibility and antibacterial	Ref
Ion-based	PAA/Gelatin/PtCNF/Ag ⁺ /DMSO	0.0212	699	-54.7 °C	1.2% (30 days)	-	-	[272]
	P(HFBA-co-HEA)/[EMIM][TFSI]	7.9×10^{-4}	992	-20–80 °C ~100% (60 days at room temperature)	√	√	-	[273]
	SF/Ca ²⁺ /DMAPS/EG	1.95×10^{-3}	>1 500	-20 °C 94% (30 days at 25 °C and 60% RH)	-	-	Cell viability: 100%	[248]
	P(AM-co-VI)/QCS/Fe ³⁺	0.038 3	633	-14 °C	-	-	-	[172]
	F127DA/SWCNT/PEDOT:PSS	4.4	>520	-	125% (~48h)	-	√	[274]
Composite-based	PAA/ChCl/Lignin-TA/graphene nanosheet	0.28	>1 800	-80–50 °C 88.56% (35 days at 25 °C and 50% RH)	-	Seamlessly repaired within 2 min at room temperature	Cell viability: ~100% <i>E. coli</i> : 93.6% <i>S. aureus</i> : 94.7%	[136]
	PAA/LM-CNC/NaCl	0.038	~2 000	-	-	90% (8 h)	√	[132]
	Gelatin/PU/PEDOT:PSS, ChCl/Gly/Borax	6.6×10^{-3}	~390	-80 °C >91.5% (30 days at 37 °C)	-	-	Cell viability: >90% <i>E. coli</i> : 83% <i>S. aureus</i> : 98.9%	[275]
	TPU/PVAc/APTES/AgNPs@MXene	√	621	-	5.5% (7 days)	-	-	[276]
	PAM/P(BA-co-HFBMA)	√	~611.3	(71.90 ± 2.55)% (7 days air-drying period)	Non-swelling state for up to 180 days	-	-	[277]
	PPy/PEDOT:PSS	8.67	-	-	~15% (250 min)	-	-	[278]
	PEDOT:PSS/ILs/Gly	~305	31	-	-	Current restored within 0.1 s	-	[279]
	PAM/PEDOT:PSS/PANi/EG/TA/TMSPMA	~24	358%	-	-	-	Cell viability:~100%	[280]

“√” stands for a mention without specific data

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4. Assembly and encapsulation design of hydrogel-based flexible sensors

Benefiting from the widespread application of advanced manufacturing techniques, such as solution processing, microstructure molding, photolithography, additive manufacturing, and fiber spinning, high-precision regulation has been achieved for the microscales and intrinsic physicochemical properties of hydrogels. However, this represents only the initial stage in the development of practical sensing technologies. The ultimate performance of hydrogel-based sensors depends not only on the physicochemical properties of the materials themselves but is also significantly constrained by subsequent device engineering design. Transforming soft, hydrated hydrogels with ionic conductivity into stable and reliable electronic devices is a systematic engineering endeavor involving multiscale structural regulation and multiphysics coordination. The core challenge lies in effectively bridging the significant disparities in mechanical modulus, conductive mechanisms, and surface energy between hydrogels and conventional electronic components such as metal electrodes, conductive wires, and encapsulation layers.

Specific device assembly strategies directly dictate the efficiency of stress transfer and the patterns of electric field distribution, serving as the key to resolving the intrinsic trade-off between sensitivity and the sensing range. Meanwhile, interface engineering acts as a bridge connecting ionic and electronic conductors, where the quality of interfacial anchoring fundamentally governs the signal-to-noise ratio (SNR) of signal transduction and the long-term stability of the device. Advanced encapsulation techniques addressing the inherent dehydration of hydrogels constitute the final safeguard for ensuring environmental adaptability. In light of this, this section systematically explores the comprehensive fabrication chain of hydrogel sensors. It begins by reviewing function-oriented device architectures and assembly strategies, proceeds to elaborate on interface design principles for achieving robust and low-impedance connections, and concludes with a discussion on encapsulation schemes for enhancing device robustness.

4.1. Assembly strategies

The assembly strategies of hydrogel-based flexible sensors serve as a pivotal bridge linking intrinsic material properties to ultimate device performance. Fundamentally, this process involves engineering geometric structures to modulate the transduction pathways from external stimuli to internal electrical signals, as well as the distribution of physical fields. Driven by variations in monitoring targets and application scenarios, current assembly strategies have evolved into five mainstream categories: vertical sandwich architecture, planar interdigital architecture, core-shell encapsulation architecture, asymmetric bilayer architecture, and monolithic architecture.

The vertical sandwich architecture encapsulates a functional hydrogel layer between two parallel electrodes (Figure 9(a)), thereby establishing a vertical current path or a normal electric field through the material. This closed, or semi-closed, layered structure is ideally suited for detecting variations in physical quantities along the thickness direction (e.g., compressive displacement under pressure) or fluctuations in bulk properties (e.g., temperature-induced changes in bulk resistance). As illustrated in Figure 9(b), this configuration can be readily extended into a sensor array via a cross-type arrangement to detect the 3D distribution of force or strain^[281]. However, owing to the presence of multiple heterogeneous interfaces, this architecture is prone to interfacial delamination. Furthermore, the mechanical modulus mismatch between the electrode layers and the hydrogel often constrains the synergistic deformation capability of the entire device^[282].

Distinct from vertical stacking architectures, the planar interdigital architecture features coplanar interdigitated electrodes constructed on a substrate via photolithography or printing technologies (Figure 9(c))^[283]. The signal transduction mechanism in this architecture is governed by planar fringing electric fields or short-range lateral conductive pathways confined within the electrode gaps^[284]. For instance, a piezoresistive sensor was constructed by placing an IAB hydrogel onto silver interdigitated electrodes^[285]. By being assembled into a 4×4 sensor array, the magnitude and spatial distribution of applied pressure could be clearly visualized (Figure 9(d)). Crucially, this “open” surface geometry not only ensures high sensitivity toward in-plane strains but also facilitates direct interaction between the sensing material and the ambient environment. Consequently, the adsorption and diffusion efficiencies of gas- or humidity-sensitive molecules are greatly improved, rendering this architecture an ideal architecture for surface-dominated sensors.

To utilize conductive hydrogels as electrodes for triboelectric nanogenerators (TENGs), a core-shell encapsulation architecture has been developed (Figure 9(e)). In this architecture, the entire device is constructed by fully sealing the hydrogel core within a dielectric elastomer shell (e.g., Ecoflex or PDMS)^[286, 287]. This topological design enables the outer elastomer to serve as the triboelectric friction layer for single-electrode mode sensing, while physical encapsulation inherently addresses the critical issues of dehydration and ion leakage. Consequently, this architecture ingeniously integrates robust self-powered capabilities with long-term environmental stability. As depicted in Figure 9(f), a triboelectric nanogenerator was fabricated by encapsulating an LM/PVA hydrogel within Ecoflex, successfully realizing self-powered wearable HMI^[288].

In the realm of epidermal electronics targeting bio-signal acquisition (e.g., ECG, EMG, and electroencephalogram (EEG)), the design priority shifts from bulk sensitivity to low-impedance interfacial

coupling. To address this, an asymmetric bilayer architecture, often featuring an anisotropic design, has been adopted (Figure 9(g)). The bottom layer comprises a hydrogel with high tissue adhesiveness and ionic conductivity, designed to establish seamless conformal contact with the skin, thereby minimizing electrode–skin interface impedance and suppressing motion artifacts^[289]. Conversely, the top layer integrates metal films or conductive fabrics as backing electrodes, facilitating signal extraction and providing mechanical support. This asymmetric architecture not only ensures mechanical compliance at the bio-interface but also effectively resolves the engineering challenge of interconnecting soft materials with external rigid circuitry. In a typical study, a high-density sEMG electrode array based on a 3D-printed polyion complex-Gly (PIC-G) hydrogel was developed^[290]. By integrating AI algorithms to decode the human hand sEMG signals, the system achieved real-time and high-precision control of a prosthetic hand (Figure 9(h)).

Furthermore, the monolithic architecture represents the most prevalent and straightforward strategy for flexible sensors (Figure 9(i)). By eschewing intricate multilayer stacking, this design utilizes the conductive hydrogel itself as the monolithic sensing body, with external circuit integration achieved via electrodes positioned at both ends. Within this framework, the sensing signals originate from the synergistic response of the bulk material. Specifically, for strain sensors, the transduction mechanism is primarily dictated by geometric deformations and the subsequent reconstruction of internal conductive networks^[291]. In the case of temperature sensors, the response stems from thermally driven fluctuations in ion mobility^[292]. This monolithic approach optimally preserves the hydrogel's mechanical properties, including its exceptional stretchability and low Young's modulus, making it highly suitable for monitoring large-scale deformations such as joint articulation. For instance, by adopting this architecture and simple terminal wiring, the developed scalable transparent sensor enables reliable, real-time monitoring of multi-site human motions and skin temperatures (Figure 9(j))^[293]. However, as the output signal reflects an integrated response of the entire bulk phase, this architecture inherently possesses limitations in spatial resolution for multi-point sensing applications.

In summary, selecting an appropriate device assembly strategy tailored to the specific requirements of sensitivity, mechanical deformability, environmental stability, and bio-interfacial interaction is a prerequisite for sensor design. Future structural designs are expected to evolve towards the integration of multiple architectures and the functionalization of micro/nanostructures.

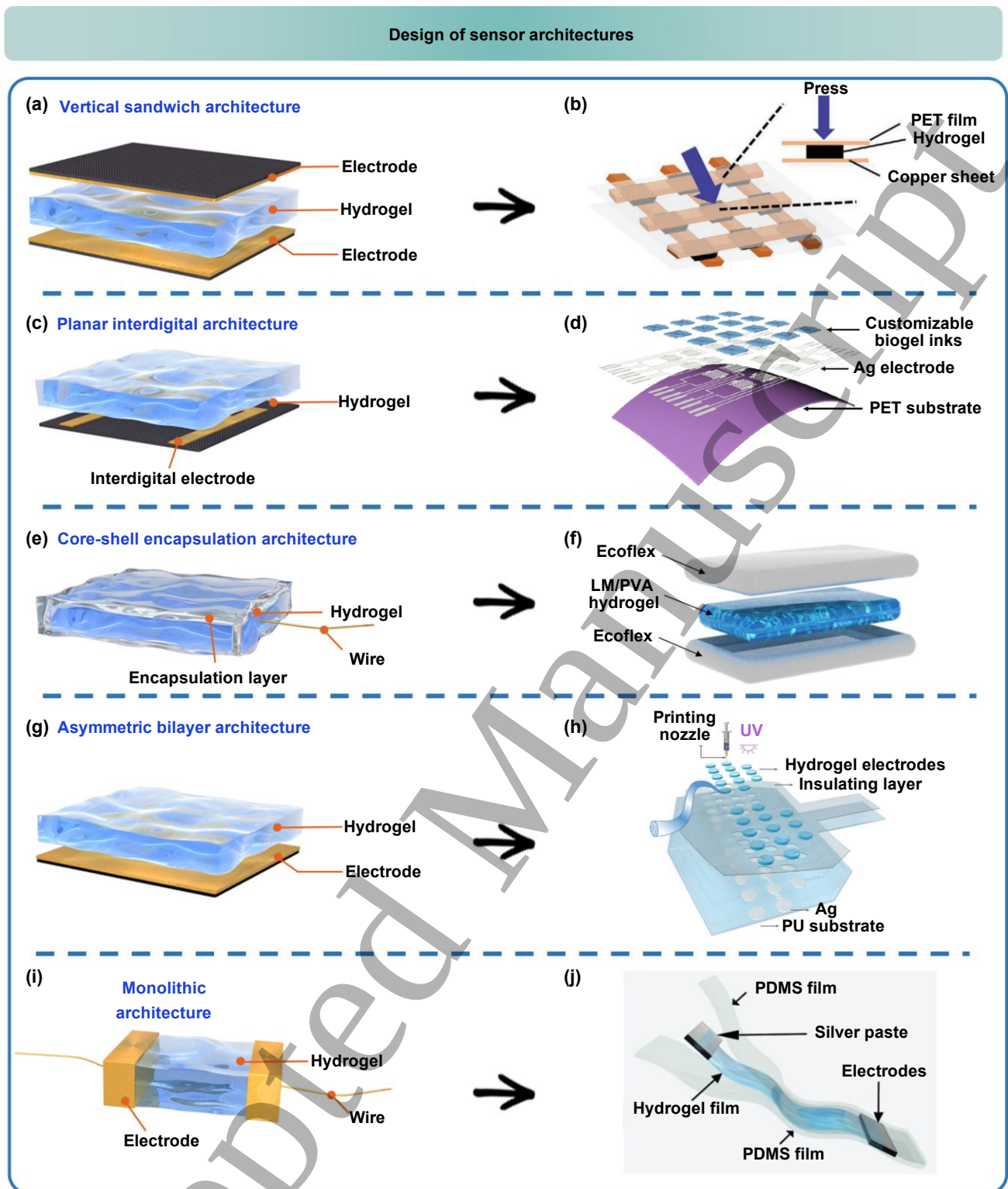


Figure 9. Assembly strategies of hydrogel-based sensors. (a) Vertical sandwich architecture. (b) Pressure sensor array using a vertical sandwich architecture. ^[281] John Wiley & Sons. © 2021 Wiley-VCH GmbH. (c) Planar interdigital architecture. (d) Piezoresistive pressure array using a planar interdigital architecture. ^[285] John Wiley & Sons. © 2025 Wiley-VCH GmbH. (e) Core-shell encapsulation architecture. (f) Ecoflex-encapsulated hydrogel for TENG. Reproduced with permission from ^[288]. © 2024 Youke Publishing Co., Ltd. (g) Asymmetric bilayer architecture. (h) The layer-by-layer architecture of sEMG electrode array. Reproduced

from ^[290]. CC BY 4.0. (i) Monolithic architecture. (j) Scheme depicting the structure of TFSS sensor. Reprinted with permission from ^[293]. Copyright (2021) American Chemical Society.

4.2. Chemical anchoring and mechanical interlocking

Within the hierarchical architecture of hydrogel-based flexible sensors, the heterogeneous interfaces where hydrogels contact metallic electrodes, elastomeric substrates, or conductive textiles serve as pivotal junctures for determining both signal integrity and long-term stability. These interfaces also constitute high-risk zones for stress concentration and failure. Owing to the substantial chemical potential difference and mechanical modulus mismatch between soft, wet hydrogels and rigid, dry electronic materials, interfaces that rely solely on van der Waals forces or simple physical adsorption are highly susceptible to delamination or slippage under dynamic cyclic loading. Such interface failure not only causes dramatic fluctuations in contact resistance, generating signal artifacts, but also disrupts ion–electron conduction pathways, leading to complete functional loss of the device. To resolve this inherent contradiction, current interface engineering strategies focus primarily on two dimensions: constructing molecular-level covalent chemical bridges and developing micro-nano-scale topological mechanical interlocking.

Covalent chemical bridging strategies aim to eliminate interfacial discontinuities at the molecular level by introducing robust chemical bonds between heterogeneous materials through surface chemical modification or in situ reactions. For instance, click chemistry principles can be utilized to establish strong interfacial chemical bonds between hydrogels and substrate surfaces^[294]. Specifically, the epoxy groups on the hydrogel surface can react with the amino groups in bovine serum albumin to form robust covalent C–N bonds. This firmly anchors conductive films to the hydrogel substrates, achieving exceptionally high adhesion strength at the interface. The resulting sensors withstand severe stretching and tens of thousands of fatigue cycles, effectively resolving interlayer separation issues. Beyond surface modification, a universal in-situ anchoring technique was developed (Figure 10(a))^[260]. Researchers subjected elastic PU porous substrates and metal electrodes to silane coupling agent self-assembly treatment, followed by in-situ polymerisation of conductive hydrogels within the pores. During polymerisation initiation, the active groups of the silane coupling agent undergo covalent cross-linking with the gel network, thereby establishing a robust integral chemical bond between the hydrogel and the device substrate.

Unlike molecular-level strategies involving chemical bonding, the topological mechanical interlocking approach focuses on leveraging the geometric confinement effects of micro- and nanostructures. By anchoring hydrogels within the pores or fiber networks of substrates, this mechanism disperses stress and impedes crack

propagation^[295]. This approach inherently circumvents chemical incompatibilities between materials, offering a universal solution for heterogeneous integration. A relatively common strategy involves permeating the uncured hydrogel precursor into the fibre network of superhydrophobic fabrics, followed by in situ thermosetting treatment (Figure 10(b))^[296]. This approach achieved structural integrity by leveraging the synergistic mechanical behaviour between the fiber network and the hydrogel matrix. For instance, robust hydrogel–elastomer hybrids are achieved through a bionic mechanically interlocked hybrid structure^[210]. Specifically, an electrospun porous thermoplastic PU fibre network serves as the interlocking scaffold, trapping the hydrogel at the microscopic scale within the elastic fibre network to form a stable interface resembling a tendon-bone connection. This strategy substantially enhances interfacial peel toughness while perfectly preserving the inherent softness and skin adhesion of the hydrogel. In summary, whether through molecular chemical bonding or microstructural interlocking, interface design is progressively evolving towards physicochemical synergistic reinforcement. This approach aims to endow flexible sensors with interface robustness surpassing that of natural tissues, enabling them to withstand extreme mechanical deformation and complex physiological environments.

Notably, although the strategies discussed above have significantly enhanced the initial adhesion strength of heterogeneous interfaces (as summarized in Table 4), their long-term stability under dynamic fatigue and extreme environments is critical for determining the operational lifetime of hydrogel-based sensors. Under dynamic fatigue, interfacial damage proceeds via crack propagation. Interfaces relying solely on chemical anchoring lack energy dissipation, causing cracks to initiate near-grafted bonds and propagate rapidly, limiting fatigue life. In contrast, interfaces with mechanical interlocking or dynamic sacrificial bonds (e.g., coordination or hydrogen bonds) dissipate energy through anchorage friction or reversible bond breakage/recombination, significantly retarding crack growth and enhancing fatigue resistance^[297]. In humid or aqueous environments, water molecules plasticize interfaces or hydrolyze susceptible bonds (e.g., ester bonds), degrading adhesion^[298]. Mitigation strategies include the use of hydrophobic barrier layers and hydrolysis-resistant bonds such as amides or coordination bonds^[299]. Under temperature fluctuations, the coefficient of thermal expansion mismatches between hydrogels and rigid substrates accumulate thermal stress, resulting in warping or delamination^[300]. Mitigation strategies include introducing modulus gradient layers or soft buffer layers to dissipate thermal stress and constructing thermally responsive, reversible bonds that release stress through dynamic exchange upon temperature changes. Understanding the origins of these

1
2 differences in long-term stability requires a fundamental examination of how these two strategies transfer
3 stress and failure.
4

5
6 Chemical anchoring and mechanical interlocking exhibit fundamental differences in stress transfer and
7 failure mechanisms. In terms of stress transfer, chemical anchoring relies on intermolecular forces (e.g.,
8 covalent or coordination bonds) across the interface. Stress is efficiently transmitted at the atomic scale
9 through these discrete bonds, enabling intimate contact without relative slippage. In contrast, mechanical
10 interlocking depends on interfacial geometric features such as microporous structures or topological
11 entanglements. Stress is transferred regionally at the micro/nano-scale through anchorage points by pushing
12 and friction, offering a buffering effect and enabling stress redistribution^[301]. Regarding failure mechanisms,
13 chemical anchoring typically leads to brittle, catastrophic failure. Once the external force exceeds the bond
14 energy threshold, irreversible bond breakage occurs, rapidly initiating crack propagation and sudden
15 interfacial debonding^[302]. Conversely, mechanical interlocking exhibits ductile, progressive failure. Under
16 stress, the anchorage points gradually deform and pull out, dissipating substantial energy through continuous
17 interfacial friction and material viscoelasticity, resulting in a slow failure process with clear precursors. This
18 fundamental distinction dictates that chemical anchoring is more suitable for high-sensitivity, small-
19 deformation sensing applications, whereas mechanical interlocking is preferable for flexible electronic devices
20 requiring high fatigue resistance, high deformation tolerance, and effective energy buffering.
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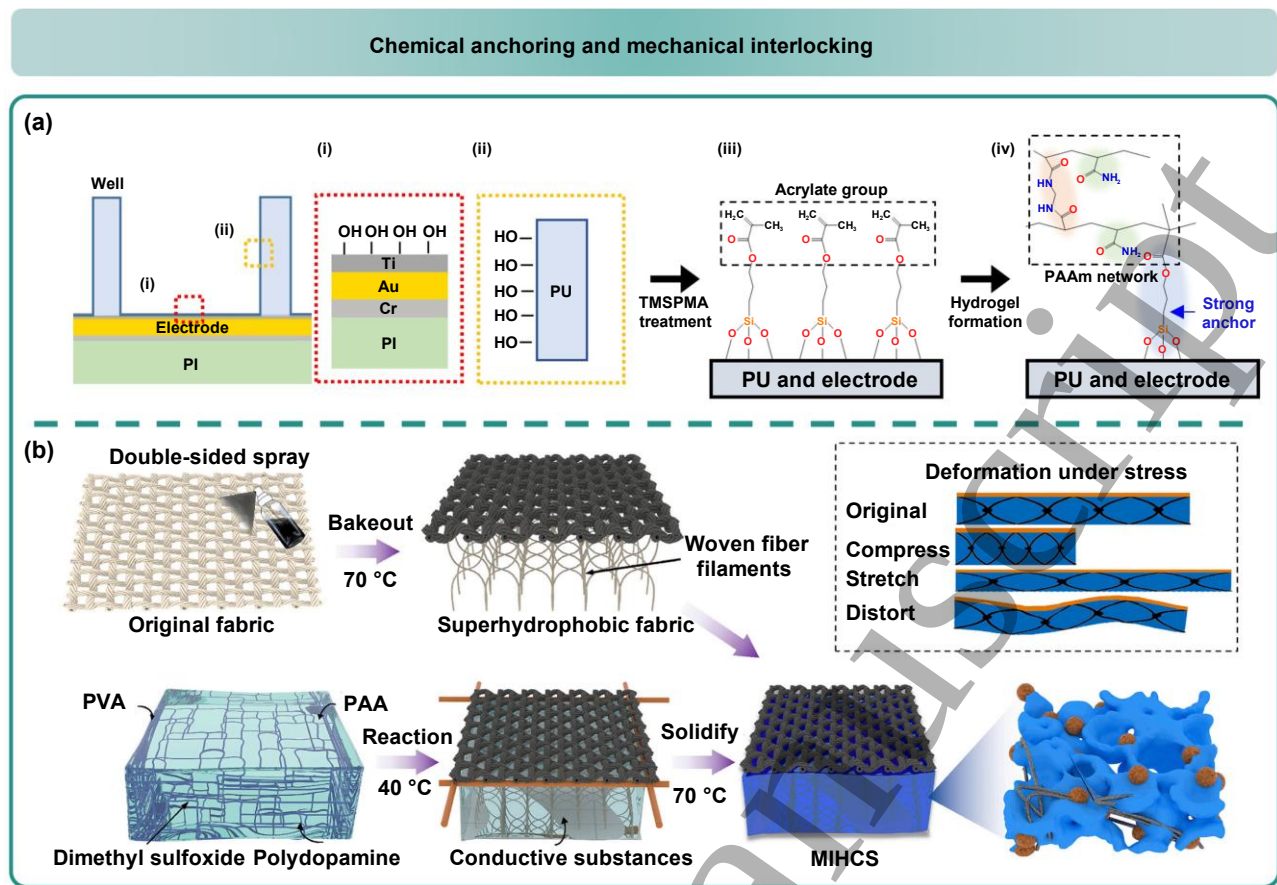


Figure 10. Chemical anchoring and mechanical interlocking strategies at heterogeneous interfaces. (a) In-situ anchoring via silane coupling for robust hydrogel–substrate interfacial bonding. Reproduced from^[260]. [260]. CC BY 4.0. (b) Fabrication process of a mechanically interlocked hydrogel–textile composite sensor. Reprinted from^[296], Copyright (2025), with permission from Elsevier.

Table 4. Representative studies on hydrogel–substrate interface combination strategies and their interfacial toughness.

Strategies	Substrate material	Hydrogel material	Specific principles	Test method	Interfacial toughness	Ref.
Chemical anchoring	AgNW/MXene/BSA film	Agar/P(AM-co-GMA)	A click chemistry reaction occurred between the amino groups contained in BSA and the epoxy groups on the gel surface, forming a C–N chemical bond	Lap-shear test	33/127 kPa	[294]
	PU	PEDOT:PSS/PANi/PAM	Chemical anchoring groups form strong covalent bonds between the conductive hydrogel and the device surfaces	90°-peeling test	72.9/279.2 J·m ⁻²	[260]
	SEBS	PDA/PAM	The interlocked hybrids are realized by infiltrating the hydrogel precursor into porous microfoam and crosslinking under ambient conditions	180°-peeling test	5.6/158.2 J·m ⁻²	[295]
Mechanical interlocking	Porous TPU	PAM/Ca-Alg	Infiltration of precursor into porous TPU webs, followed by thermal curing	90°-peeling test	3.56/50.9 J·m ⁻²	[210]
	Zn	PAM/CAR/Gly	The hydrogel and the electrode undergo a micron-scale interpenetration through electrochemical reactions	180°-peeling test	10/87 J·m ⁻²	[303]

“Interfacial toughness” refers to the interfacial toughness before and after modification

4.3. Encapsulation strategies for environmental resistance

Although material modification strategies, such as the incorporation of organic solvents or ILs, as discussed in Section 3.2, have significantly enhanced the intrinsic environmental tolerance of hydrogels, relying solely on the stability of the material matrix often proves insufficient to fully withstand the erosion of extreme environments in practical, long-term service scenarios. For instance, ILs may exhibit minimal exudation during prolonged use, potentially triggering tissue inflammation^[304]. Similarly, highly hygroscopic salts can still absorb moisture gradually under extremely high humidity, leading to dimensional instability of the device^[305]. Therefore, from a device engineering perspective, constructing an external encapsulation system capable of effectively and physically blocking the cross-interface transport of water molecules stands as a primary prerequisite for ensuring the reliable functionality of hydrogel-based electronic devices in complex and dynamic environments.

Current mainstream encapsulation strategies primarily revolve around establishing physical barriers. Elastomer encapsulation is a widely adopted method, whose core mechanism involves constructing a dense physical barrier around hydrogels using hydrophobic polymers such as silicone rubber or Ecoflex. This locks

the ionic concentration and swelling state of the hydrogel within a stable range. However, simple physical encapsulation often leads to layer delamination due to incompatibility at the hydrophilic–hydrophobic interface. To address this challenge, the incorporation of amphiphilic stearic acid as a molecular bridge to physically link hydrogel fibres with externally encapsulated silicone has been demonstrated to be a viable strategy (see Figure 11(a))^[306]. Leveraging its amphiphilic properties, this created robust physical anchoring at the originally incompatible hydrogel–elastomer interface, achieving outstanding dimensional stability. Moreover, more robust interfacial chemical cross-linking strategies are frequently employed to achieve stable connections between encapsulated elastomers and hydrogels. For instance, benzophenone is utilised to initiate covalent bonds between the hydrogel surface and elastomer groups^[307]. This ‘welded’ interface not only endows the device with exceptional deformation resistance but also significantly enhances the water vapour barrier efficiency through the formed covalent bond network.

While static encapsulation enhances stability, conventional elastomer coatings inevitably develop microcracks under prolonged repeated stretching, leading to barrier failure. Inspired by the dynamic properties of biological membranes, dynamic adaptive coatings offer a more sophisticated solution to fatigue failure under deformation (Figure 11(b))^[308]. This strategy mimics the fluid-mosaic model of cell membranes, centred on constructing a dynamic, self-repairing protective layer. Within the coating, long-chain amphiphilic molecules anchor to the hydrogel surface like membrane proteins, while the overlying lipid molecular layer forms a dense hydrophobic barrier. The most ingenious design feature lies in preloading excess lipid molecules within the hydrogel matrix. When surface coatings develop microscopic defects due to device deformation, such as stretching or bending, these internal lipids migrate to the surface like repair molecules, dynamically filling voids and restoring barrier integrity. This adaptive repair mechanism ensures sustained prevention of water evaporation under both static and dynamic conditions, while maximally preserving the inherent flexibility, conductivity, and even biocompatibility of the hydrogel. This offers a novel pathway towards achieving long-term, reliable biointegrated electronic devices.

In summary, from developing elastomer encapsulation that forms static physical barriers to advancing biomimetic coatings that provide dynamic adaptive protection, researchers are systematically addressing the environmental stability challenges of hydrogel devices through multi-level mechanism innovations. These strategies significantly expand their application potential in complex scenarios such as wearable electronics, soft robotics, and biomedicine.

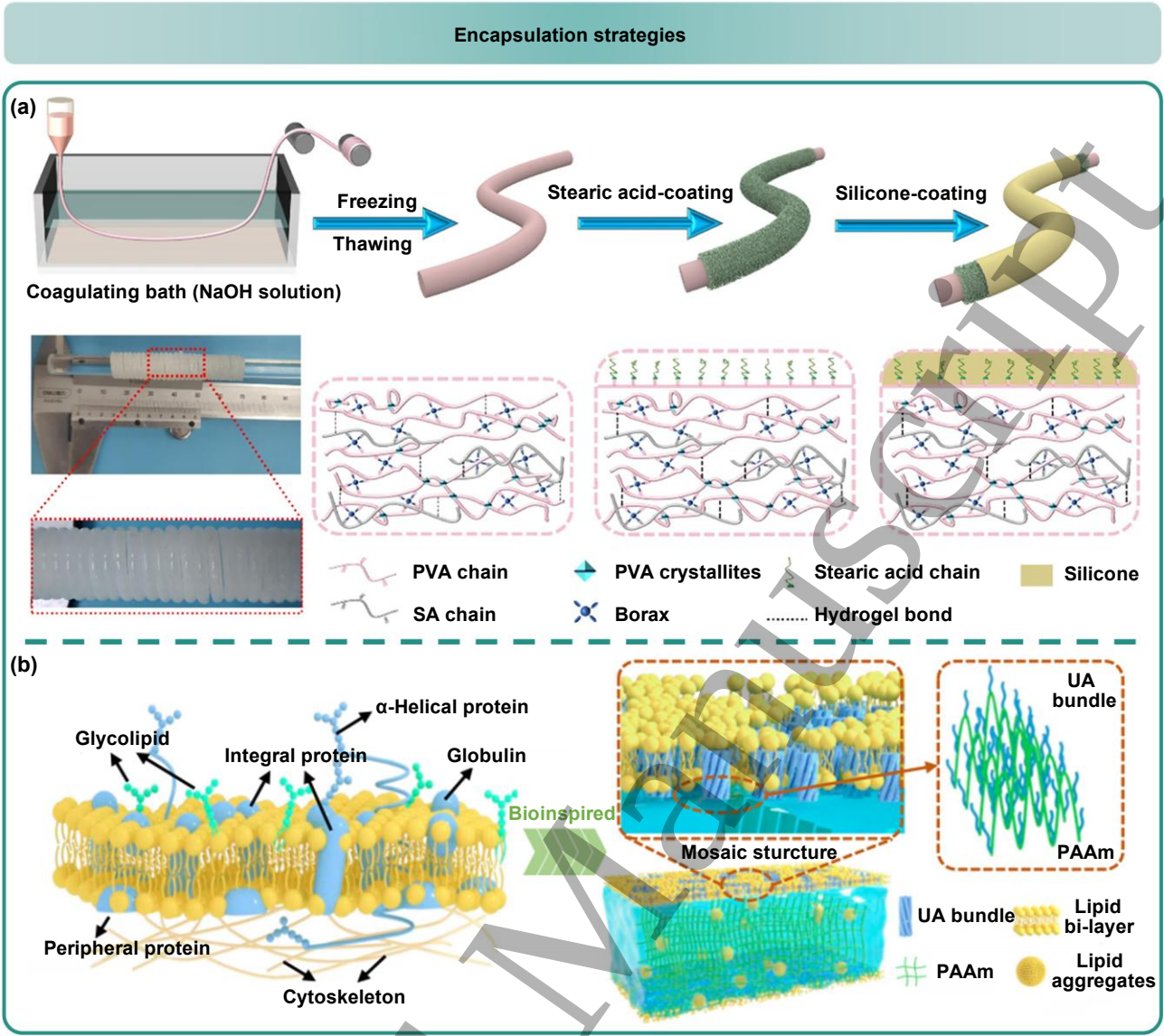


Figure 11. Encapsulation strategies for hydrogel-based sensors. (a) The preparation process of the silicone rubber-encapsulated hydrogel fibers with enhanced interfacial bonding. Reprinted from [306], Copyright (2024), with permission from Elsevier. (b) Biomimetic dynamic adaptive coating for self-repairing hydrophobic barrier. Reproduced from [308], with permission from Springer Nature.

4.4. Challenges in interface engineering and encapsulation strategies

4.4.1 Interfacial failure mechanisms under multi-field coupling. The failure mechanisms of heterogeneous interfaces under multi-field coupling service conditions remain poorly understood. Current reliability assessments are largely limited to single-dimensional mechanical tests. There is a lack of qualitative and quantitative understanding of the interfacial physicochemical evolution under coupled factors such as dynamic fatigue, environmental stresses (e.g., osmotic swelling pressure, thermomechanical strain), and chemical erosion. This knowledge gap makes it difficult to accurately predict the long-term service life of devices in complex, dynamic real-world scenarios. Future research should employ in-situ, multi-modal characterization

techniques to reveal the accumulation and evolution mechanisms of interfacial damage. Furthermore, the development of physical failure models that describe the coupled mechanothermal–electrochemical interactions at the interface is urgently needed^[309].

4.4.2 Electrochemical instability of ion–electron interfaces. The electrochemical stability of ion–electron conducting interfaces requires urgent improvement. During long-term cyclic deformation, microscopic perturbations in the interfacial contact state can cause drastic drift in contact impedance. This induces significant signal artifacts and baseline instability, severely compromising the fidelity of sensing signals. Meanwhile, the high-concentration free ion environment within hydrogels poses a continuous electrochemical corrosion risk to metal electrodes, leading to physical fracture and functional loss of conductive pathways^[310]. Therefore, developing novel conductive interfacial layers that combine high mechanical compliance, strong interfacial adhesion, and electrochemical inertness (such as introducing two-dimensional materials, CPs) is critical for ensuring long-term device stability^[311].

4.4.3 The functional paradox of encapsulation systems. The encapsulation system presents a prominent functional paradox and design imbalance. The high dependence of hydrogels on moisture retention and the rigid demand for physiological breathability in wearable applications constitute a fundamental performance paradox. Existing encapsulation systems, while pursuing efficient prevention of water loss, often completely block necessary gas and vapor exchange. This results in a functional design imbalance that not only severely limits wearing comfort and physiological compatibility but also may cause skin inflammation due to localized microenvironmental changes. The development of anisotropic or responsive encapsulation materials is essential to achieve synergistic optimization of moisture retention and breathability^[307].

5. Application in flexible electronics

The material synthesis strategies and device assembly architectures detailed in the preceding sections establish a robust structural foundation for hydrogel-based sensors. However, the efficacy of these systems hinges on their capacity to efficiently transduce multidimensional physical, chemical, or physiological stimuli into quantifiable electrical signals. Despite the burgeoning diversity of application scenarios, the underlying signal transduction mechanisms primarily operate via five fundamental principles (as illustrated in Figure 12): resistive, capacitive, triboelectric, piezoelectric, and bioelectric coupling. Each of these sensing mechanisms exhibit distinct characteristics. The resistive mechanism centers on the modulation of charge carrier transport pathways. External stimuli can alter the geometric topology of the conductive network within the gel or affect ionic mobility, with these microscopic changes ultimately manifesting as variations in overall resistance^[49].

1
2 The capacitive mechanism originates from dynamic changes in charge storage capability, reflected either in
3 the spatial redistribution of dielectric polarization or the reorganization of the electrical double layer at the
4 electrode interface^[81]. The triboelectric mechanism is built upon the synergistic coupling of contact
5 electrification and electrostatic induction. When materials with different electronegativities come into contact,
6 electron or ion transfer occurs at the interface, and the resulting displacement current drives carrier flow in the
7 external circuit^[312]. The piezoelectric mechanism requires a non-centrosymmetric structure, fundamentally
8 involving the reorientation of electric dipoles under mechanical stress. As internal polarization undergoes
9 transient changes in response to stress, a macroscopic potential difference emerges^[313]. The bioelectrical
10 coupling mechanism focuses on the transduction process between ionic and electronic signals across interfaces.
11 Benefiting from the high water content of hydrogels and their structural similarity to biological tissues, this
12 mechanism achieves high-fidelity conversion of physiological bioelectrical signals into electronic currents in
13 artificial circuits through ultra-low interfacial impedance^[314].

14
15 Capitalizing on these mechanisms and leveraging their tissue-mimicking modulus and superior
16 biocompatibility, hydrogels not only resolve the mechanical mismatch at the biotic–abiotic interface but also
17 serve as a pivotal interactive medium bridging the physical world with digital systems. In this section, the
18 diverse applications of hydrogel sensors are categorized into four primary domains: pressure sensors, which
19 enhance force perception precision via modulus regulation and energy dissipation; strain sensors, which
20 achieve broader detection ranges and higher sensitivities through hydrogel integration; and environmental
21 sensors, which utilize stimuli-responsiveness to efficiently monitor temperature, humidity, and pH; and
22 bioelectric sensors, which exploit the ultra-low impedance at the hydrogel–tissue interface for high-fidelity
23 electrophysiological acquisition. Table 5 provides a comparative summary of the performance metrics of
24 recent hydrogel-based flexible sensors. The incorporation of hydrogels not only circumvents the limitations
25 of traditional materials but also paves new avenues for the intelligence, multifunctionality, and bio-integration
26 of flexible sensors.

27
28 The achievement of the aforementioned performance relies on precise control over the microstructure of
29 hydrogel-based sensors, which underscores the importance of the fabrication techniques detailed in Section 2.
30 Solution processing facilitates scalable platform fabrication for diverse sensor types through accurate
31 regulation of film thickness and uniformity. Microstructure molding and photolithography construct surface
32 topologies that create stress concentration regions in pressure sensors to boost sensitivity^[315], while also
33 generating rough surfaces for environmental or bioelectronic sensors to increase the contact area with the

external environment^[316]. Additive manufacturing exploits design freedom in 3D geometry to introduce buffer structures in strain sensors for expanded detection ranges^[317] and to engineer complex architectures in pressure sensors, enabling high-sensitivity linear response^[318]. Fiber spinning yields one-dimensional continuous structures ideal for integrating strain sensors with stretchable textiles, and its core-sheath configurations support the development of multimodal sensing fibers^[319]. In essence, the fabrication strategies outlined in Section 2 endow hydrogel-based sensors with diverse structural characteristics, critically shaping their performance across applications in pressure and strain sensing, environmental monitoring, and bioelectrical recording.

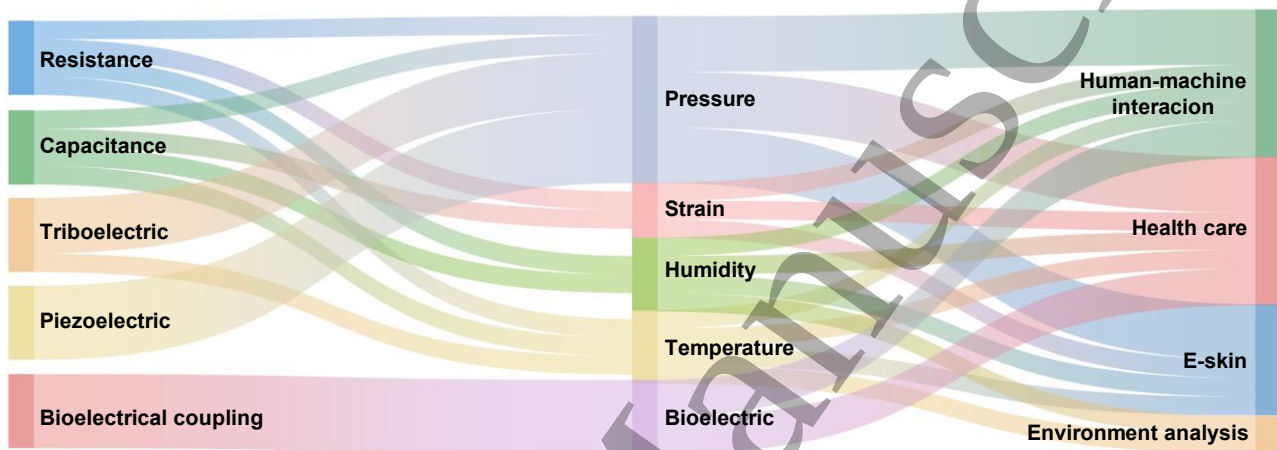


Figure 12. Connections between sensing mechanisms, perception dimensions and application scenarios.

Table 5. Comparison of the sensing performance of flexible sensors based on different types of hydrogels.

Component	Mechanism	Type	Sensitivity	Response / Recover time	Sensory application	Ref.
MXene/Fmoc-FF	Piezoresistive	Pressure	38.5 kPa ⁻¹ (0–400 kPa)	0.64 ms / 0.61 ms	Sound monitoring	[320]
PAM/AG/CaCl ₂	Piezocapacitive	Pressure	1.96 kPa ⁻¹ (0–25 kPa) 7.56 kPa ⁻¹ (25–100 kPa)	30 ms / 15 ms	Smart glove operating	[4]
Sulfonated lignin-Fe ³⁺ /AM/DMAPS	Piezocapacitive	Pressure	65.82 kPa ⁻¹ (~0.024–200 kPa)	112.6 ms / 225.2 ms	Motion monitoring	[321]
PVA/SA/PANI	Triboelectric	Pressure	2.29 kPa ⁻¹ (5–40 kPa)	46 ms / 40 ms	Drone operations	[43]
Polyphenols (SL)-Fe ³⁺ /P(HEA-co-AA)/LiCl	Triboelectric	Pressure	2.641 kPa ⁻¹ (0.016–0.16 kPa)	41 ms / 30 ms	Human motion monitoring	[322]
			0.112 kPa ⁻¹ (0.16–3.2 kPa) 0.005 kPa ⁻¹ (3.2–48 kPa)			
Gel/OCS-ABTO	Piezoelectric	Pressure	49.61 mV·kPa ⁻¹ (0–1.27 kPa)	24 ms / 21 ms	Health monitoring	[323]
PAAN/Gly/Zn ²⁺	Piezoelectric	Pressure	1.34 mV·kPa ⁻¹ (0–176.84 kPa)	31 ms / 51 ms	Human motion monitoring and human-machine interaction	[324]
PVA-CE	Piezoionic	Pressure	1 490 nV·Pa ⁻¹ (0–1 kPa)	18.1 ms	E-skins	[325]
			63 nV·Pa ⁻¹ (1–10 kPa) 5 nV·Pa ⁻¹ (10–100 kPa)			
PAP-MCP	Piezoionic	Pressure	7.2 mV·kPa ⁻¹ (0.11–50 kPa) 3.46 mV·kPa ⁻¹ (50–180 kPa)	66 ms / 98 ms	Human-computer interaction	[326]
PVA/PAM	Piezoresistive	Pressure	0.05 kPa ⁻¹ (below 3.27 kPa) 0.02 kPa ⁻¹ (3.27–6.83 kPa)	150 ms	Human motion monitoring	[327]
HEA/EOEOEA/LiCl/TPO	Piezocapacitive	Pressure	171 kPa ⁻¹ (0–60 kPa)	5.6 ms	Health monitoring	[328]
			12 kPa ⁻¹ (60–250 kPa)			
CMC/P(AA-co-DAC)/NaCl/LM	Piezoelectric	Pressure	0.268 kPa ⁻¹ (0.016–1.6 kPa)	35 ms / 30 ms	Respiratory signal monitoring	[329]
			0.008 kPa ⁻¹ (1.6–56 kPa)			
P(AA-co-NASC)/PVDF-TrFE/ CaCl ₂ /DMSO	Resistive	Strain	0.6 (0%–100%)	-	Underwater soft gripper and underwater signal monitoring	[108]
			1.1 (100%–200%)			
			1.3 (200%–400%)			

Table 5. Continued.

Component	Mechanism	Type	Sensitivity	Response / Recover time	Sensory application	Ref.
PVA-PPy@CNF-RPc	Resistive	Strain	1.92 (0%–100% $\perp$) 0.066 (0%–100% //)	0.3 s / 0.4 s	Human motion monitoring and robot arm control	[330]
PVA/PAA/borax	Capacitive	Strain	0.8% (0%–100%)	-	Physiological signals and human motion monitoring	[331]
HK/PAM/Mxene	Resistive	Strain	1.79 (0%–100%) 5.27 (100%–500%) 10.22 (500%–1 000%)	130 ms / 200 ms	Health monitoring	[332]
PAA/HEMA@nanoclay	Resistive	Strain	0.75 (0%–400% //) 0.57 (0%–400% $\perp$)	50 ms	Human motion monitoring	[333]
PAA/GT/CNTs/Zr ⁴⁺	Resistive	Strain	2% (0%–600%)	70 ms / 120 ms	Underwater motion monitoring and communication	[334]
PVA/PAM/Gly/MBA/TMEDA	Resistive	Temperature	-1.137 5%·°C ⁻¹ (20–70 °C) -1.027 91%·°C ⁻¹ (70–80 °C)	-	Alarm system and temperature detection	[335]
PBA/CPA/Gly	Resistive	Temperature	-0.020 1%·°C ⁻¹ (20–80 °C)	-	Localized temperature detection	[336]
PAM/SA	Resistive	Temperature	Change magnitude ~38.8% (22–100 °C)	2.02 s	Health monitoring	[337]
PAM/Tapioca	Capacitive	Humidity	13 462.1%·%RH ⁻¹ (11%–98% RH)	276 s / 227 s	Respiratory monitoring	[68]
PVDF/LiCl	Capacitive	Humidity	12 678.96 pF·%RH ⁻¹ (25%–95% RH)	17.7 s / 21 s	Respiratory monitoring	[338]
SA/Br	Capacitive	Humidity	9% RH ⁻¹ (5%–30%) 235% RH ⁻¹ (30%–70%) 1 580% RH ⁻¹ (70%–100%)	0.9 s / 0.5 s	Detection of subtle humidity differences	[339]

5.1. Pressure sensors

Pressure sensors, as a critical branch in the field of flexible sensors, have garnered extensive attention and research. Based on their working mechanisms, hydrogel-based pressure sensors can be further classified into piezoresistive^[320, 340–343], piezocapacitive^[4, 321, 344, 345], triboelectric^[43, 139, 322, 346], piezoelectric^[323, 324, 347], and piezoionic sensors^[325, 326, 348–350].

Hydrogels exhibit exceptional flexibility and stretchability, enabling them to accommodate complex deformation environments. This characteristic allows hydrogel-based stress sensors to conform to intricate surfaces (such as human skin or mechanical joints) and undergo uniform deformation under external forces, thereby achieving high-precision detection of stress distribution^[351]. The combination of rigid conductive polymer networks with entangled soft polymer chains was implemented to establish mechanically robust hydrogels^[327]. Specifically, AM and PVA were selected to construct a dual-network architecture crosslinked by MBA. As shown in Figure 13(a), the hydrogel-based pressure sensors exhibited reliable piezoresistive behavior under dynamic pressures (3.33, 5.02, and 6.67 kPa) and were employed to monitor diverse human physiological activities, such as vocalization, swallowing processes, finger movements, and limb activities.

Ionic hydrogels have played a pivotal role in flexible pressure sensors due to their high ionic conductivity, flexibility, self-healing properties, and biocompatibility. When a hydrogel-based pressure sensor is subjected to external pressure, the 3D crosslinked network of the hydrogel undergoes compression. Such structural modifications significantly shorten ion migration pathways and increase local electric field intensity, thereby inducing alterations in electrical performance. During this process, the electric double layer (EDL) mechanism plays a pivotal role^[328, 352–354]. When the hydrogel interfaces with electrodes, an EDL structure spontaneously forms at the contact boundary, wherein free ions within the hydrogel redistribute under electric fields to establish Stern and diffuse layers. Upon pressure application, hydrogel compression modulates the charge distribution profiles, consequently inducing interfacial capacitance/resistance variations. This EDL-governed electromechanical coupling enables ionic hydrogel pressure sensors to achieve pressure detection with enhanced sensitivity and rapid response time, simultaneously unveiling novel pathways for developing high-performance flexible electronics with biomimetic sensing capabilities. For instance, a micro-dome dielectric hydrogel (LSHM) utilising a sulphonated lignin (LS)-Fe³⁺ autocatalytic system was employed to construct a capacitive pressure sensor (SCPS)^[321]. Benefiting from the LSHM's outstanding mechanical properties and elastic recovery rate, the SCPS demonstrated a broad detection range and high sensitivity. As illustrated in Figure 13(b), its working principle relied on the EDL capacitance formed between the electrodes and the

dielectric layer. When pressure was applied, the contact area between the dielectric layer and the electrodes increased, resulting in a capacitance change. In volleyball training, SCPS assisted athletes in optimizing their movements through pressure mapping.

Additionally, incorporating piezoelectric materials can confer piezoelectric properties to hydrogels. As shown in Figure 13(c), a PAAN/Gly/Zn²⁺ piezoelectric organic hydrogel was constructed by incorporating polyvinylidene fluoride (PVDF), a common piezoelectric material^[324]. PAAN represents the abbreviation for polyacrylonitrile, acrylonitrile (AN), AM, and p-styrenesulfonate (NaSS). This organohydrogel integrated piezoelectricity, low-temperature tolerance, mechanical robustness, and stable electrical properties. Researchers further developed wearable sensors and electronic gloves based on organohydrogels to monitor diverse human activities, achieving sensitive detection of finger bending, elbow flexion, vocalization, pulse beats, and gestures.

Human mechanosensation relies on ionic currents regulated by cutaneous mechanoreceptors^[355, 356]. Inspired by the sensory behavior of skin, piezionic sensors have been developed^[357]. Hydrogel-based piezionic sensors transduce mechanical stimuli into electrical signals through internal ion migration. Specifically, applied pressure causes asymmetric redistribution of mobile ions (e.g., Na⁺ and Cl⁻) within the hydrated polymer network, generating diffusion potential between electrodes. Inspired by spider webs, researchers developed an aluminium-coordinated hydrogel (HG-Al_{PAC})^[349]. It was composed of polymers such as PVA, PAA, and PAM, along with Al³⁺ ions and the metal-ligand ions [(OH)Cl₂]³⁻, forming a spider-web-like structure. This hydrogel showed excellent mechanical properties, including a tensile toughness of 2.75 MJ·m⁻³, and good anti-fatigue and anti-puncture abilities. The piezionic sensor based on HG-Al_{PAC} worked by promoting Cl⁻ separation and migration under pressure, converting mechanical stimuli into electrical signals (Figure 13(d)). It exhibited a high piezionic coefficient of 0.89 mV·kPa⁻¹ and an energy conversion efficiency of 1.29%, with good stability and self-powered characteristics. This sensor could be used for human motion monitoring and served as a tactile nerve system for analgesic patients to prevent injuries.

Hydrogels, characterized by their high water content and excellent ionic conductivity, play a pivotal role in triboelectric nanogenerators by effectively facilitating charge generation and transport^[329]. In a typical configuration, the hydrogel is encapsulated within an insulating elastomer, serving as an electrode. When an external object separates from the elastomer surface, residual static charges on the elastomer induce the directional migration of ions within the hydrogel to neutralize the interfacial charges, resulting in an accumulated excess of ions at the interface. Simultaneously, an electrical double layer forms and polarizes at

1
2 the interface between the metal wire and the ionic hydrogel, leading to an accumulation of negative ions at
3 the interface and corresponding positive charges in the metal wire. To equilibrate this double-layer structure,
4 electrons flow from the metal wire through the external circuit to the ground until all static charges within the
5 elastomer film are fully screened. For instance, a single-electrode sandwich structure PPG-TENG was
6 constructed using a multifunctional dual-network hydrogel based on PAM, PDA, and Go via silicone rubber
7 (SR) encapsulation^[139]. As shown in Figure 13(e), the mechanism relies on contact electrification and
8 electrostatic induction effects. As the dielectric material came into proximity with the SR layer, charge
9 separation occurred at the interface, creating opposing charge distributions. During the separation process, the
10 resulting positive charges in the hydrogel layer induced electrons through the external circuit to the ground,
11 thereby generating an electrical pulse signal. The PPG-TENG delivered high electrical outputs (143 V open-
12 circuit voltage (OCA), 1.8 μ A short-circuit current, and 47 nC transferred charge), responded quickly, and
13 could detect forces as low as 0.1 N, making it suitable for monitoring subtle human movements like facial
14 expressions and breathing.

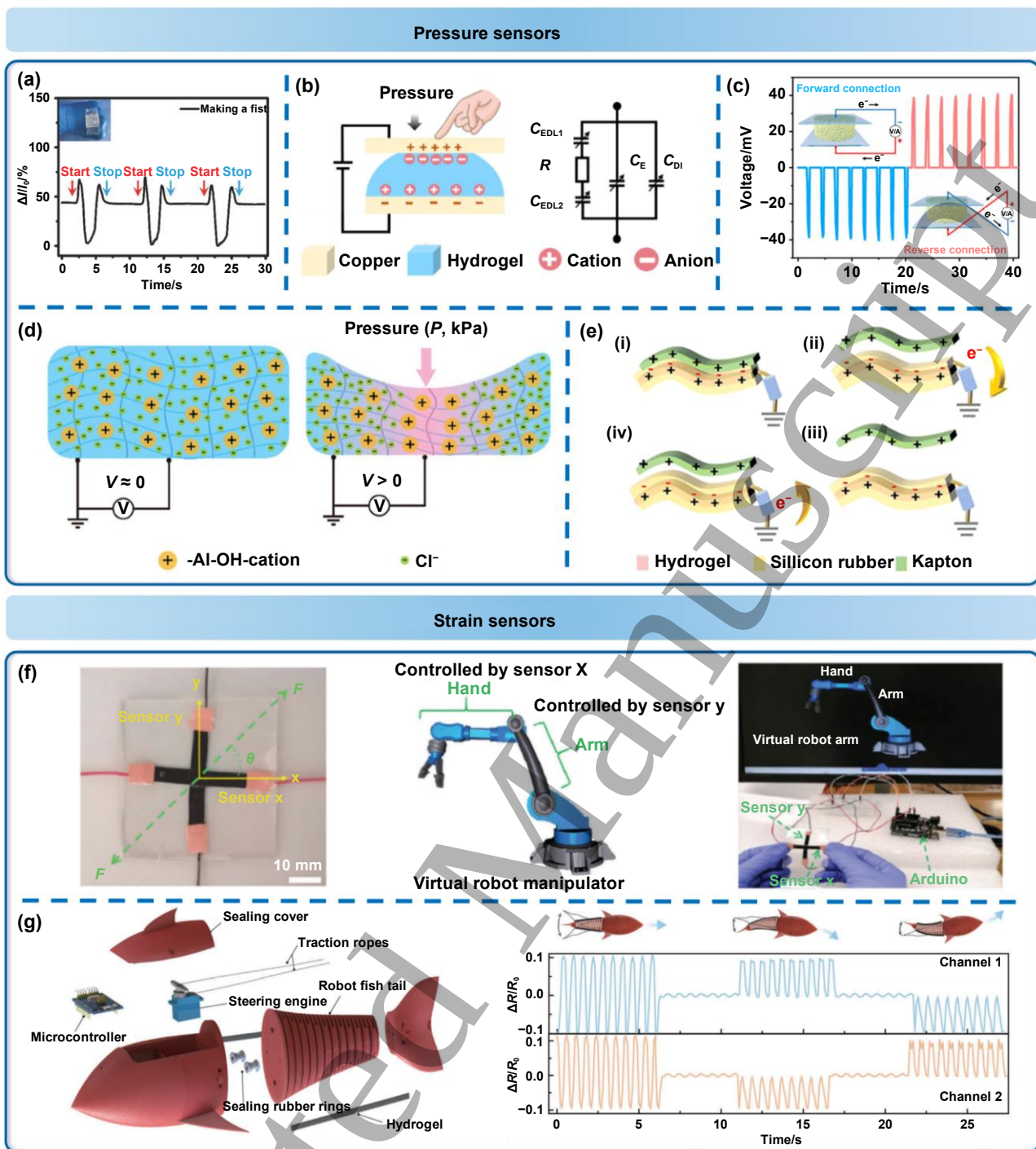


Figure 13. Multifunctional hydrogels for flexible sensors. (a) Grip signals recorded by hydrogel-based piezoresistive sensors. [327] John Wiley & Sons. © 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (b) The sensing mechanism of SCPS based on the EDL effect. Reprinted from [321], Copyright (2025), with permission from Elsevier. (c) Generated output voltages of PAAN/Gly/ Zn^{2+} organohydrogel-based PENG during forward and reverse connections. Reprinted with permission from [324]. Copyright (2024) American Chemical Society. (d) Schematic of the piezoionic effect of HG-AIPAC. [349] John Wiley & Sons. © 2024 Wiley-VCH GmbH. (e) Schematic diagram of the sensing mechanism of the PPG-TENG. Reproduced from [139] with permission from the Royal Society of Chemistry. (f) Photograph of a cross-plied PVA-PPy@CNF5-RPc100 hydrogel strain sensor and its virtual robot manipulator controlled via a microcontroller. Reproduced with permission from [330]. CC BY-NC 4.0. (g)

Structure of an underwater fish-imitating robot and resistance change curves of hydrogel-based strain sensors on both sides of the robot under different modes of swimming. [358] John Wiley & Sons. © 2023 Wiley-VCH GmbH.

5.2. Strain sensors

Hydrogels have been extensively investigated and established as premier candidates for high-performance flexible strain sensors^[197, 331, 332]. Operating primarily through piezoresistive or capacitive mechanisms, these sensors transduce mechanical stimuli into electrical signals via geometric deformation or the reconfiguration of internal conductive networks. Compared with traditional rigid sensors limited by narrow strain ranges (<5%) and dry elastomer-based devices prone to interfacial noise, hydrogels demonstrate superior sensing capabilities. The unique solid–liquid hybrid state endows these sensors with an ultra-wide linear detection range (often exceeding 1 000%) and exceptional fracture resistance, enabling comprehensive monitoring from subtle pulse vibrations to large-scale joint movements. Meanwhile, the ionic transport mechanism circumvents the percolation threshold failures common in dry conductive composites, ensuring superior signal linearity and stability even under extreme deformation. Furthermore, the ultra-soft, conformal interface eliminates contact resistance fluctuations (motion artifacts) during dynamic activities, significantly enhancing the SNR and the limit of detection for minute stimuli.

To maintain signal stability and linearity over ultra-large strain ranges, ionic conductive hydrogels based on a nanoclay physical interlocking mechanism have been developed^[333]. This nano-reinforced interpenetrating network endows the material with superior structural–mechanical–electrical integrity, allowing the internal conductive pathways to reorganize stably even under substantial deformation. Consequently, high-fidelity and reliable sensing performance is achieved. Leveraging this robustness, the sensor can intimately conform to the dynamically beating heart surface, enabling not only the sensitive capture of strain signals but also the accurate localization of myocardial infarction areas, demonstrating significant potential for monitoring pathological physiological changes.

Traditional conductive hydrogels typically exhibit intrinsic isotropy in their structure and function, as they are synthesized through the homogeneous polymerization of monomers. This structural characteristic sharply contrasts with the highly ordered anisotropic architecture observed in biological soft tissues (e.g., muscles, tendons, and cartilage). To simulate this anisotropic structure, an anisotropic conductive hydrogel with a direction-oriented polymer network was constructed by crosslinking PVA, PPy-decorated cellulose nanofibres, and TA^[330]. The hydrogel exhibited outstanding mechanical performance, with a tensile strength

of 11.41 MPa and a toughness of $12.44 \text{ MJ} \cdot \text{m}^{-3}$. Additionally, it displayed anisotropic adhesion behavior and direction-dependent conductivity. As shown in Figure 13(f), hydrogel strain sensors utilizing these properties were fabricated to exhibit superior sensitivity in multiple directions. This innovative sensor design allowed for reliable tracking of complex human joint motions with multiple degrees of freedom while simultaneously achieving the control of a multi-axis virtual robot manipulator.

Operational stability under long-term and high-frequency cycling remains a critical challenge. To address this, a conductive PEDOT:PSS-PVA hydrogel was fabricated via directional freeze-casting and salting-out treatments^[358]. This material combines a low modulus ($\sim 100 \text{ kPa}$) with high extensibility ($>600\%$) and exceptional fatigue resistance ($>300 \text{ J} \cdot \text{m}^{-2}$). Such robust mechanical properties stem from an optimized microstructure, which effectively dissipates external stress and inhibits crack propagation. When applied as an underwater sensor, the device monitored rapid fish tail oscillations with high signal consistency. Notably, it maintained a stable performance without mechanical failure for more than 100 000 loading cycles (Figure 13(g)). These results strongly validate the feasibility of reliable monitoring in harsh dynamic environments.

5.3. Environmental sensors

Hydrogels exhibit significant potential in environmental sensing due to the efficient detection of changes in parameters such as temperature, humidity, and pH. These unique stimulus-responsive characteristics and multifunctional integration make such materials ideal for environmental monitoring.

Temperature, as a fundamental physical parameter reflecting the thermal state of matter, plays a critical role in human health monitoring, industrial production, and environmental control. For instance, abnormal body temperature serves as an early indicator of numerous diseases, while abnormal equipment temperature may indicate operational failures or safety hazards. To achieve precise temperature monitoring, hydrogel-based temperature sensors have emerged^[336, 337]. These sensors leverage the high sensitivity of hydrogel materials to external temperature variations. Through either the thermal motion of internal water molecules or phase-transition behaviors of polymer networks, they convert temperature changes into electrical signals, enabling real-time temperature detection. For instance, thermosensitive and self-adhesive PEST hydrogels were employed in highly sensitive temperature sensors^[359]. As shown in Figure 14(a), this wearable sensor displayed a good linear response between 20°C and 80°C . It could distinguish temperature changes at different body parts (as small as 1°C) and monitor temperature fluctuations caused by motion in real time (to detect a 20°C difference).

Humidity, as another critical physical parameter in environmental monitoring, exerts extensive influence

across industrial production, agricultural management, healthcare, and daily life. For instance, excessive humidity in industrial settings may accelerate equipment corrosion, while precise humidity control in medical environments directly impacts patient comfort and infection prevention. To enable high-precision real-time humidity monitoring, hydrogel-based humidity sensors have become a research focus^[338]. These sensors exploit the inherent hygroscopicity of hydrogels, where moisture absorption-induced polymer network swelling or ionic conductivity variations convert humidity changes into electrical signals, achieving accurate humidity quantification. As a representative example, an organic hydrogel with a κ -carrageenan/PAM dual-network structure was employed for humidity sensing^[360]. The sensor achieved a rapid 0.27 s response and 0.3 s recovery times across a broad range of 4%–90% relative humidity (RH). The sensor's mechanism relied on changes in ionic conductivity due to varying humidity levels: when the RH is low, a decreased water content decreases KCl solubility and ion movement, thus increasing resistance; conversely, when the RH is high, greater water absorption facilitates ion dissociation and transport, thereby lowering resistance (Figure 14(b))^[360].

pH, as a critical parameter for assessing solution acidity/alkalinity, plays a vital role in biomedical research, environmental monitoring, food processing, and chemical production. For instance, minute pH variations in human blood may indicate severe health conditions, whereas abnormal pH levels in aquatic systems can disrupt ecological balance. Hydrogel-based pH sensors have emerged as a research priority to achieve high-precision real-time pH monitoring. These sensors utilize the pH-dependent responsiveness of hydrogel materials. The protonation/deprotonation reactions of functional groups (e.g., $-\text{COOH}$ or $-\text{NH}_2$) under various pH values induce polymer network swelling or contraction, thereby converting changes in the pH into electrical signals for precise detection. As a representative demonstration, high-selectivity, pH-responsive hydrogels were developed by functionalizing a PEG polymer backbone with pH-sensitive tertiary amine groups^[361]. As shown in Figure 14(c), this material was further integrated into a wireless, bioabsorbable pH sensor for early anastomotic leakage detection post-gastrectomy. Acidic environments caused by gastric leakage triggered water and proton diffusion into the hydrogel, resulting in volumetric expansion. This swelling altered the inductance of the embedded LC circuit, which could be precisely quantified via inductive coupling with external reader devices.

Gas detection is important in environmental monitoring, industrial safety, medical diagnostics, and daily life. For instance, toxic gas leaks may endanger lives, while volatile organic compound detection aids in air quality assessment and disease diagnosis. Current research has achieved gas-sensing technologies for

1 ammonia^[362, 363], nitrogen dioxide^[364], O₂^[307, 365], and others^[366, 367], using hydrogels as gas-sensing materials.
2
3 The inherent flexibility and elasticity of hydrogels ensure user comfort, enabling real-time, reliable monitoring
4 of target gas concentrations at ambient temperature, thereby providing long-term protection for health and
5 safety. For instance, a dual-network hydrogel (DNH) composed of PAM and calcium alginate (CA) was
6 employed as the electrolyte in a self-powered, battery-structured gas sensor for H₂S monitoring^[366]. The
7 electrode potential in such galvanic cells is fundamentally influenced by the interaction between the electrode
8 materials and the electrolyte. The dynamic response curves of Zn/Ag/DNH, Fe/Ag/DNH, and Zn/Fe/DNH
9 toward H₂S are presented in Figure 14(d). It was revealed that sensors containing Ag electrodes exhibited
10 substantial variations in open circuit voltage (OCV), indicating a chemical interaction between Ag and H₂S
11 that altered the electrode surface composition and reduced the OCV. In practical applications, this sensor
12 successfully achieved non-invasive halitosis diagnosis and meat freshness monitoring. When integrated with
13 Bluetooth and cloud technologies, it further enabled remote H₂S monitoring, demonstrating potential
14 applications in portable medical electronics and food safety applications.
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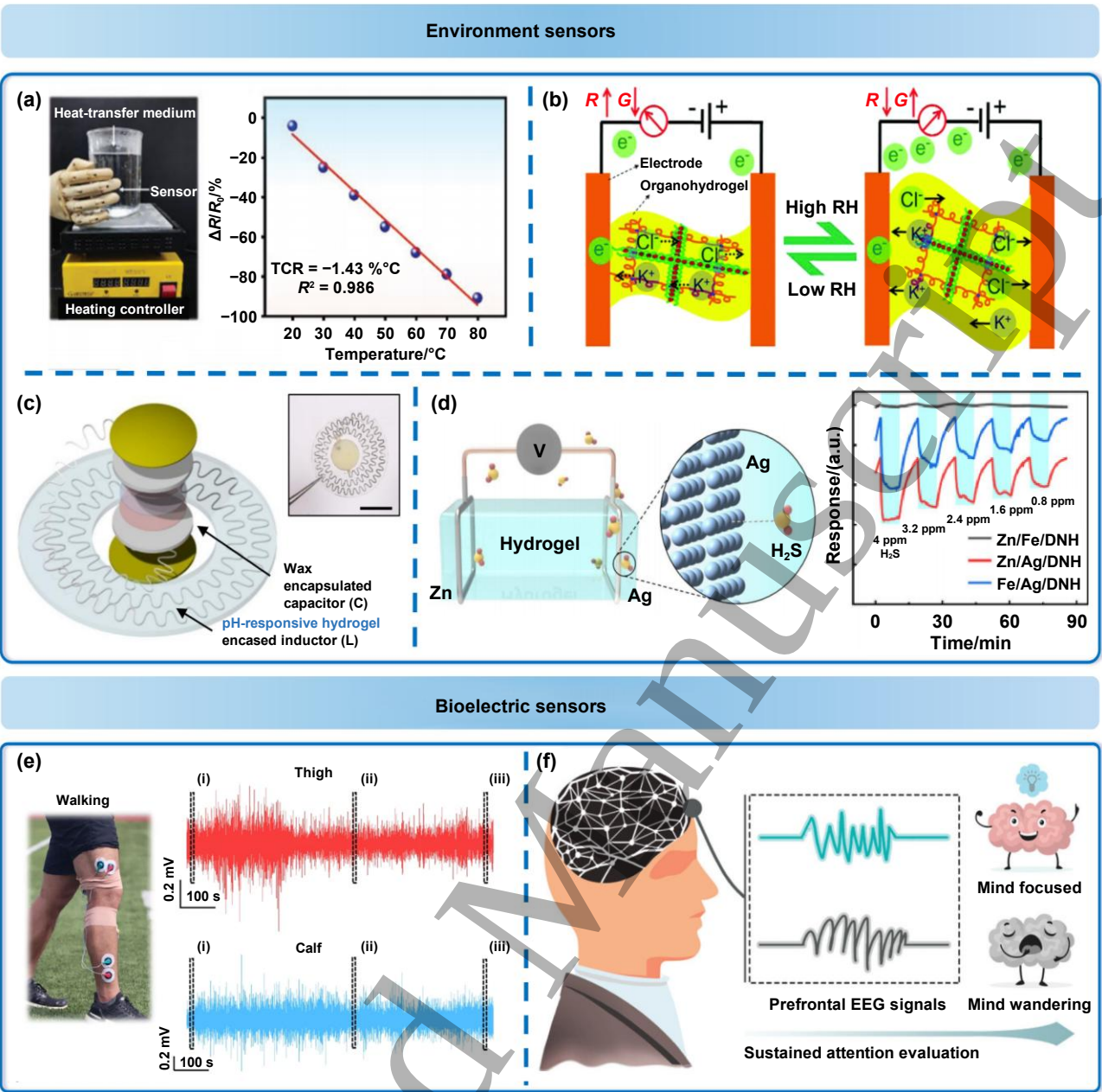


Figure 14. Multifunctional hydrogels for environmental sensors and bioelectric sensors. (a) Schematic diagram of a hydrogel thermistor epidermal sensor attached to a prosthetic hand to measure temperature and its temperature response curve. [359] John Wiley & Sons. © 2023 Wiley-VCH GmbH. (b) The principle of humidity sensing in organohydrogel. Reproduced from [360] with permission from the Royal Society of Chemistry. (c) A wireless bioresorbable pH sensor using a pH-responsive hydrogel. Reproduced from [361]. CC BY 4.0. (d) Structural configuration and H_2S detection mechanism of the Zn/Ag/DNH gas sensor, with comparative dynamic response analysis of different architectures to stepped H_2S concentrations (4–0.8 ppm). Reproduced from [366]. CC BY 4.0. (e) sEMG signals of the thigh and calf during walking. Reproduced from [368], with permission from Springer Nature. (f) The hydrogel electrodes for acquiring EEG signals to assess sustained attention. [34] John Wiley & Sons. © 2023 Wiley-VCH GmbH.

5.4. Bioelectric sensors

Hydrogels have emerged as ideal candidates for flexible bioelectrode materials due to their inherent flexibility, biocompatibility, and adhesiveness. The low modulus enables conformal adhesion between flexible electrodes and skin, mitigating biomechanical mismatch and relative motion at the electrode–tissue interface, thereby ensuring stable signal acquisition. The exceptional biocompatibility prevents skin irritation or damage during prolonged wear, making them suitable for medical health monitoring. The low interfacial impedance facilitates high-accuracy transfer of weak physiological signals from tissues to electrodes with enhanced signal-to-noise ratios. Meanwhile, their strong adhesion allows stable tissue to contact without additional adhesives, effectively suppressing interference from extreme deformation, motion, or vibration while improving wearing comfort. Furthermore, high transparency enables convenient electrode manipulation and long-term observation while guaranteeing biosafety. These properties have garnered increasing attention for constructing seamless and conformal skin–electrode interfaces. To date, numerous hydrogel-based electrodes have been implemented for detecting human physiological and biochemical information, including ECG^[369, 370], EEG^[371], EMG^[36], and electrooculogram (EOG)^[372].

ECG signal, a physiological indicator of heart function, results from the heart's electrical activity. Originating from the sinoatrial node, atrial muscle, atrioventricular node, and His-Purkinje system, this electrical activity regulates the heart's rhythmic contractions and relaxations. In clinical settings, ECG monitoring is vital for diagnosing cardiovascular issues such as arrhythmia, myocardial ischemia, and infarctions, as it can detect abnormalities in waveforms such as the P wave, QRS complex, and T wave. Beyond diagnosis, ECG monitoring is also valuable for heart surgery, rehabilitation, and exercise physiology studies, offering insights into heart function, guiding exercise regimens, and tracking cardiac health. For instance, researchers have developed an ionic hydrogel from PVA, branched polyethyleneimine, and CaCl₂ to record ECG signals^[35]. This non-covalently cross-linked network hydrogel featured 1291% stretchability, 3.09 S·m⁻¹ conductivity, strong adhesion, antibacterial properties, and recyclability. The temperature-sensitive adhesion enhanced skin contact, reducing interfacial impedance. As a flexible electrode, it could record ECG, EMG, and EEG signals with high fidelity, high signal-to-noise ratio, and ultralow detection limits, showing promise in healthcare electronics. For instance, the system could distinguish ECG signal differences between male and female subjects during rest and post-exercise conditions using epidermal electrodes.

EMG signals are critical bioelectrical indicators reflecting muscle activity and neural control states. By detecting weak electrical signals generated by muscle fibers during contraction, EMG provides clinicians with

direct insights into muscular functional status and neural conduction pathways. In fields such as sports medicine, rehabilitation therapy, and occupational disease prevention, EMG technology demonstrates exceptional utility, enabling professionals to assess muscle fatigue levels, diagnose neuromuscular disorders, and guide athletes in scientific training protocols. For instance, an in-situ biogel composed of gelatin, PEDOT:PSS, and deep eutectic solvents was developed to enable rapid liquid-to-solid transformation upon skin contact. As shown in Figure 14(e) of this work, the biogel-integrated wireless electronic system was applied to record sEMG signals from the thigh and calf muscles during a 15-minute walking session, enabling the estimation of walking frequency and assessment of muscle conditions through curve signals and spectrograms^[368].

EEG signals reflect electrical potential variations corresponding to cerebral and emotional state changes. Consequently, EEG signals can aid in the diagnosis of Parkinson's disease, Alzheimer's disease, and epilepsy, as well as identifying depression and infantile autism, while also enabling emotion assessment and sustained attention evaluation. For instance, quantifying sustained attention levels during prolonged demanding tasks in aviation, surgical operations, and military missions is critical for accident mitigation and operational efficiency/accuracy enhancement. A hydrogel network composed of PVA and polyvinylpyrrolidone (PVP), which exhibits a tissue-like modulus and exceptional flexibility, has been employed for monitoring EEGs^[34]. PDA nanoparticles (PDA-NPs) obtained through PDA peroxidation were incorporated into the hydrogel, endowing it with enhanced transparency, superior adhesive properties, and reduced interfacial impedance. Based on these advantages, the multi-channel hydrogel electrode achieved conformal contact and stable sensing with the skin. In sustained attention assessment tests utilizing prefrontal EEG signals, the hydrogel electrode was able to accurately (91.5%) differentiate between seven levels of attention, highlighting its significant potential for attention-state recognition applications (Figure 14(f)).

5.5. Challenges in the practical application of hydrogel-based sensors

5.5.1. Long-term stability. At the sensor level, long-term unreliability, manifested as signal baseline drift and dynamic response distortion, represents a primary bottleneck for practical applications. First, the stability of the electrochemical interface under bias voltage remains unclear. During prolonged or dynamic loading, the electrical double layer at the electrode/gel interface is susceptible to ion depletion^[373] or electrochemical reactions^[303]. These processes cause interfacial impedance drift, leading to baseline fluctuations and a reduced signal-to-noise ratio. Second, the intrinsic viscoelasticity of hydrogels induces mechanical hysteresis, which compromises dynamic response fidelity. Such hysteretic behavior not only distorts dynamic signal waveforms

but also prevents the sensor from returning to its initial state after complex deformation histories, thereby introducing cumulative measurement errors^[204]. For applications requiring long-term continuous monitoring, signal drift and dynamic distortion arising from electrochemical instability and viscoelasticity constitute the core challenges in validating the reliability of hydrogel-based sensors.

5.5.2. In-vivo biocompatibility. In implantable applications, long-term biocompatibility and functional stability constitute the most formidable barriers to clinical translation. First, the foreign body reaction triggered by the implant, including non-specific protein absorption and macrophage accumulation, can isolate the sensor from the physiological environment for material exchange, leading to signal drift or even complete failure^[374]. Second, for hydrogels that are degradable or contain nanofillers (such as CPs or metal nanowires), their degradation kinetics and the long-term biological effects of their metabolic byproducts remain unclear^[375]. The long-term retention or release of these nanofillers in the body may induce chronic inflammation or unforeseen cytotoxicity. Therefore, achieving long-term stable operation of implantable sensors requires not only that the material itself possesses bioinert or controlled degradation properties but also the design of anti-fouling surfaces, controlled drug-release interfaces, and encapsulation strategies that ensure long-term sealing at the device level.

5.5.3. Multi-signal crosstalk. Signal crosstalk under multi-field coupling and environmental interference significantly compromises the measurement accuracy and selectivity of hydrogel-based sensors. These sensors typically transduce physical (pressure, strain, and temperature) or chemical (humidity, ions, and biomolecules) stimuli into resistance or capacitance changes within their bulk phase or at interfaces. However, hydrogels are inherently sensitive to environmental fluctuations, making target signals susceptible to interference from non-specific physical fields^[376]. This multi-physics coupling effect is further amplified in complex real-world environments, such as the surface of human skin. Therefore, addressing this challenge requires a dual approach. The first involves material design, such as introducing functional components insensitive to specific stimuli^[377]. The second relies on signal processing, including the development of multi-parameter synchronous acquisition and decoupling algorithms^[378]. Achieving effective shielding or active compensation of environmental noise, such as temperature and humidity variations, is essential for enhancing the detection accuracy of hydrogel-based sensors in practical scenarios.

1
2 **6. Conclusions and perspectives**
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4 This review establishes a comprehensive framework spanning fabrication methodologies, performance
5 regulation, assembly design, and device applications to systematically summarize the research progress on
6 hydrogel-based flexible sensors. First, advanced manufacturing processes, including solution processing,
7 microstructure molding, photolithography, additive manufacturing, and fiber spinning, are elucidated as the
8 cornerstones for realizing device structure customization. Furthermore, feasible strategies are revealed for
9 constructing efficient conductive networks, extreme environmental adaptability, such as anti-freezing and
10 water retention, self-healing capabilities, and biocompatibility, along with antibacterial properties. To address
11 the integration leap from pure materials to high-performance devices, this review summarizes chemical
12 anchoring interconnection techniques at heterogeneous interfaces and advanced encapsulation strategies.
13 These are key engineering pathways to solve device-level delamination failure and guarantee mechanical
14 integrity and environmental resistance. The deep integration of these materials and engineering technologies
15 collectively endows hydrogels with broad application potential in fields such as pressure and strain sensing,
16 environmental sensing, and bioelectric sensing. Finally, looking ahead, as shown in Figure 15, despite
17 significant progress in existing research, the following challenges remain to be addressed in this field:
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31 **(1) Signal decoupling and interference elimination**
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33 A major core challenge faced by hydrogel-based sensors in practical applications is their inherent
34 multidimensional cross-sensitivity. This frequently results in complex signal crosstalk (particularly baseline
35 drift induced by temperature and humidity) and pronounced viscoelastic hysteresis effects, thereby reducing
36 measurement reliability. To address this challenge, existing solutions primarily fall into two categories: passive
37 compensation and active decoupling. In terms of material and structural design, researchers aim to achieve
38 specific responses to individual physical quantities. For instance, composite materials with opposite thermal
39 response characteristics are employed to offset temperature interference^[379] or strain-isolation structures such
40 as island-bridge configurations are used to eliminate coupling from mechanical signals^[380, 381]. In back-end
41 signal processing, AI algorithms demonstrate strong decoupling potential, enabling feature extraction from
42 mixed signals and enabling independent quantification of multimodal stimuli^[382]. However, most current
43 decoupling techniques can only distinguish between two types of stimuli, making them inadequate for dealing
44 with the complex multi-physics interference present in real-world environments. The highly specific ion
45 transport mechanisms of biological sensory receptors (such as mechano-gated and thermosensitive channels
46 in the skin) provide key inspiration for next-generation decoupling strategies. By constructing anisotropic
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hydrogel networks with stimulus-responsive gating mechanisms, devices can be designed to open ion transport pathways only under specific physical fields, such as pressure or certain temperature thresholds. The deep integration of such intrinsic selective materials with back-end intelligent algorithms holds promise for eliminating interference at its source, achieving high-dimensional, low-energy, and crosstalk-free sensing akin to biological systems.

(2) Large-scale fabrication and standardization

Current laboratory-scale technologies (such as molding and single-nozzle printing) struggle to meet the industrial's stringent demands for high throughput, cost-effectiveness, and device consistency. Transitioning from laboratory prototypes to factory-ready products requires not only the adoption of continuous manufacturing processes like roll-to-roll processing or high-speed screen printing, but also an urgent need for innovation in the manufacturing paradigm itself. With respect to the preparation strategy, the traditional trial-and-error approach to formulation exploration is costly and inefficient. The introduction of emerging tools such as AI is poised to reshape this process, steering hydrogel preparation strategies towards a data-driven transformation. Through machine learning algorithms, researchers can swiftly screen vast datasets to identify the optimal precursor formulations for desired performance combinations^[383]. Second, in device fabrication, developing integrated multi-material continuous manufacturing technologies is key to avoiding interface defects caused by stepwise assembly. Continuous multi-material integrated 3D printing based on DIW has been developed for the in-situ heterogeneous integration of functional hydrogels, conductive electrodes, and encapsulation materials^[384, 385]. Notably, the rheological properties of hydrogel inks are highly sensitive to environmental conditions, serving as a critical factor in determining printability^[386]. Future research should extend beyond automation with fixed parameters and focus on incorporating feedback loops enabled by computer vision and machine learning^[387]. By developing intelligent closed-loop control systems capable of monitoring ink status in real time and dynamically adjusting printing parameters, rigid open-loop processes can be replaced, fundamentally addressing batch-to-batch variability and ensuring the yield and reliability of large-scale production.

(3) System-level closed-loop integration and energy autonomy

To overcome the reliance on bulky external power supplies and computational units, hydrogel-based sensors must evolve from passive sensing terminals to autonomous intelligent systems capable of "sensing-decision-feedback" functionality. Regarding energy and self-driven sensing, although self-powered hydrogel devices based on TENG/PENG have achieved preliminary integration of energy and sensing^[388–390], their

pulsed output and high internal resistance characteristics constrain the system's sustained operational capability. Inspired by the mechanically gated ion migration mechanisms within skin mechanoreceptors, the emerging piezoionic effect offers a highly promising new pathway^[350, 391]. This quasi-DC output characteristic, which is based on ion migration, not only enables more stable self-driven sensing but also has potential for enhancing energy harvesting and management efficiency through coupling with double-layer capacitance. In terms of information processing, while embedded AI and edge computing enable low-latency data processing^[392], the significant modulus mismatch between conventional silicon-based circuits and soft hydrogels remains the foremost obstacle to integration. A more forward-looking approach involves establishing neuromorphic computing architectures based on ion-electronics. By constructing artificial synapses or ionic logic gates, signals can be processed and stored in situ directly using ion flows within the hydrogel^[393]. This bio-inspired paradigm of synchronous sensing-computation is expected to fundamentally eliminate the energy-consumption bottleneck of analog-to-digital conversion, thereby contributing to the development of highly integrated, wireless, and untethered next-generation soft intelligent terminals.

(4) Device-level reliability under dynamic service

Under dynamic service conditions, a significant gap remains between material stability (e.g., resistance to dehydration) and device-level reliability. To address signal distortions caused by mechanisms such as mechanical fatigue, electrochemical polarization drift, and interfacial delamination, a paradigm shift toward comprehensive reliability assurance is imperative. At the material level, constructing highly dissipative, fatigue-resistant networks is the foundational step. Researchers have identified multiple implementation strategies, such as utilizing sacrificial bonds within multi-network structures to dissipate energy through bond rupture^[394] or incorporating porous frameworks (e.g., covalent organic frameworks, COFs) to form molecular-scale mechanically interlocked networks, where the confined sliding of polymer chains within nanochannels enables efficient energy dissipation^[395]. A more forward-looking approach involves integrating dynamic covalent bonds or supramolecular interactions to achieve in situ self-healing^[396], allowing devices to autonomously repair micro-cracks at their inception, thereby fundamentally combating mechanical fatigue. At the interface level, integrated construction strategies based on topological interlocking or modulus gradient designs offer a viable path to mitigate delamination during dynamic operation^[397]. Concurrently, replacing purely electric double-layer capacitance with Faradaic processes^[398] can establish non-polarized interfaces, thereby suppressing potential drift during long-term operation. Finally, in terms of evaluation, there is an urgent need to establish accelerated aging test standards that account for coupled multi-physics, ensuring that

devices maintain high-fidelity signals and near-zero baseline drift over tens of thousands of dynamic cycles.

In summary, this review systematically outlines the panoramic technical landscape of hydrogel-based flexible sensors, ranging from material structure–property relationships to device integration applications. Owing to their unique bio-interface compatibility and multimodal sensing capabilities, hydrogels are breaking the physical barriers between biotic and abiotic systems, reshaping the technological boundaries of flexible electronics. It is expected that this review will stimulate deep synergy across materials science, microelectronics engineering, and biomedicine, accelerating the transition of hydrogel electronics from laboratory prototypes to next-generation sensing systems characterized by industrial-grade reliability and closed-loop intelligence.

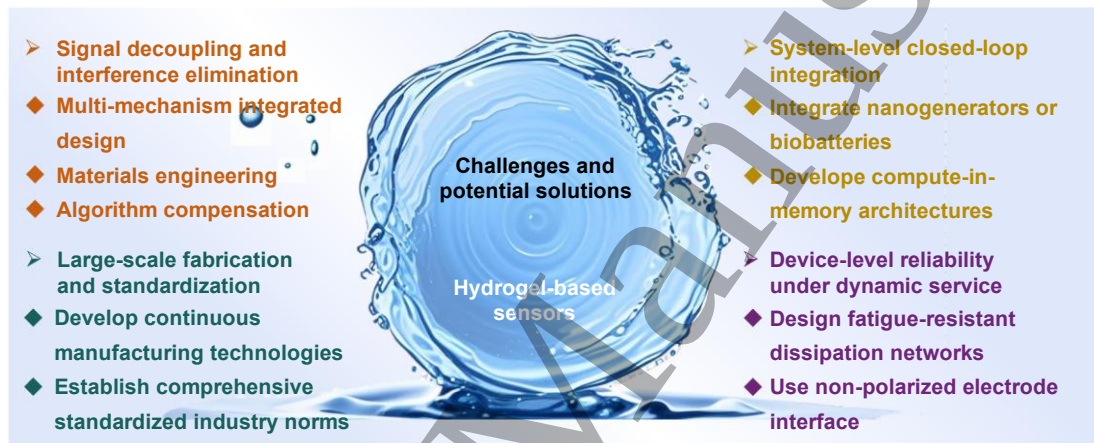


Figure 15. Current challenges and potential solution strategies for hydrogel-based sensors.

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Conflict of Interest

The authors declare that they have no conflicts of interest.

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